

Comprehensive Project Report: Type I Interferon Immunotherapy Response

Single-Cell RNA-Seq Analysis and Clinical Comparisons

Course/Project Title: Type I Interferon Immunotherapy Response: Single-Cell RNA-Seq

Analysis Prepared for: Student Project Guide (Project 3)

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Abstract

This project replicates and critically evaluates the single-cell RNA-sequencing (scRNA-seq) pipeline modeling patient responses to anti-PD-1 immunotherapy based on baseline Type I Interferon (IFN-I) response capacity (IRC). Using a dataset of peripheral blood mononuclear cells (PBMCs) from healthy donors and cancer patients, we constructed a single-cell atlas comprising 94,513 cells. We calculated an individual-cell IRC module score using a six-gene interferon-stimulated protein (ISP) signature consisting of BST2, EIF2AK2, ISG15, MX1, IFIT3, and IRF7. Patient-derived cells were stratified into IRC-High and IRC-Low states using a cell-level median threshold, followed by differential expression and over-representation analysis (ORA) using GSEApv. Our computational pipeline successfully identified distinct transcriptional states corresponding to hyper- and hypo-interferon responsiveness. Comparing our findings with the primary literature (Boukhaled et al., Nature Immunology, 2022) revealed a critical clinical alignment: pre-therapy IFN-I priming of peripheral effector T cells (Teff) serves as an epigenetic poising mechanism that drives T-cell exhaustion and limits therapeutic response, while interferon resistance predefines longer overall survival.

1. Introduction

Type I interferons (IFN-I), including IFN- α and IFN- β , are central regulators of innate and adaptive immune responses and play an important role in anti-tumor immunity. IFN-I signaling can promote antigen presentation, immune-cell activation, and expansion of tumor-reactive lymphocytes, but persistent or dysregulated IFN-I signaling can also contribute to immune dysfunction and therapeutic resistance. This dual role creates an important biological paradox in cancer immunology: IFN-I activity can support anti-tumor responses while also contributing to T-cell exhaustion and immunosuppressive feedback.

Immune checkpoint blockade targeting programmed cell death protein 1 (PD1) has improved outcomes in several malignancies, yet only a subset of patients experience durable clinical benefit. Understanding the pre-existing immune states that influence responsiveness to anti-PD1 therapy is therefore important for identifying biomarkers of treatment outcome and for clarifying the mechanisms that determine therapeutic success or failure.

The original study by Boukhaled et al. investigated whether pre-therapy responsiveness to IFN-I in circulating immune cells could predict outcome following PD1 blockade. Using single-cell mass cytometry, the authors quantified IFN-I-stimulated protein expression across peripheral immune populations and developed an IFN-I Response Capacity (IRC) score based on six core interferon-

stimulated proteins: BST2, PKR/EIF2AK2, ISG15, MX1, IFIT3, and IRF7. In the original study, IRC was defined from the change in expression of these proteins between IFN- β -stimulated and unstimulated cells.

A central finding of the study was that lower pre-therapy IFN-I responsiveness in CD4 and CD8 effector T cells was associated with improved survival after anti-PD1 therapy, whereas high pre-existing IRC was linked to disease progression and poorer outcome. The study further demonstrated that high- and low-IRC states were associated with distinct transcriptional and epigenetic programs, suggesting that IFN-I responsiveness is not simply a consequence of baseline interferon-stimulated gene expression but reflects a broader pre-programmed cellular state.

The present project used single-cell RNA-sequencing data from two healthy donors and eight cancer patients to computationally explore related IFN-I-associated transcriptional states. The analysis included quality control, doublet removal, normalization, highly variable gene selection, dimensionality reduction, Leiden clustering, canonical immune-lineage marker assessment, IRC-related module scoring, differential-expression analysis, and pathway enrichment.

Because the available dataset did not contain the paired IFN- β -stimulated and unstimulated measurements used in the original study, the present analysis does not reproduce the original IRC calculation exactly. Instead, a cell-level module score based on the same six-gene signature was calculated from baseline normalized gene expression. The objective was therefore to evaluate whether these genes define a coherent interferon-associated transcriptional state in PBMCs and whether IRC-High and IRC-Low cells differ in broader gene-expression and pathway-level programs.

The specific aims of the project were to:

- construct and quality-control a single-cell PBMC atlas across healthy donors and cancer patients;
- identify major circulating immune lineages using canonical marker genes;
- quantify cell-level heterogeneity in the six-gene IFN-I-associated signature;
- compare transcriptional profiles of IRC-High and IRC-Low patient-derived cells;
- identify biological pathways enriched in the IRC-High state; and
- compare the resulting transcriptional patterns with the biological conclusions of the original study.

2. Methods

The single-cell RNA-sequencing workflow was implemented in Python using Scanpy and AnnData. Data loading, quality control, doublet detection, normalization, highly variable gene selection, dimensionality reduction, clustering, interferon-response scoring, differential expression, and pathway

enrichment were performed sequentially. Intermediate AnnData objects, quality-control summaries, diagnostic figures, and analysis tables were saved throughout the workflow to support reproducibility and independent review.

2.1 Data Loading and Quality Control

Individual 10x Genomics PBMC count matrices from two healthy donors (HD1 and HD2) and eight cancer patients (P1–P8) were loaded using `sc.read_10x_mtx()`. Sample identity was recorded in `adata.obs["sample"]`, and donor type was assigned according to sample prefix: samples beginning with HD were classified as healthy donors, whereas samples beginning with P were classified as patients.

The ten datasets were concatenated into a single AnnData object. The merged raw dataset contained 101,020 cells and 36,601 genes.

Three quality-control gene categories were defined using gene-name prefixes:

- **Mitochondrial genes:** genes beginning with MT -
- **Ribosomal genes:** genes beginning with RPS or RPL
- **Hemoglobin genes:** genes beginning with HBA, HBB, HBD, HBE, HBG, HBM, HBQ, or HBZ

Cell-level quality-control metrics were calculated using `sc.pp.calculate_qc_metrics()`.

Cells were initially filtered using the following thresholds:

- detected genes per cell (`n_genes_by_counts`) ≥ 300
- mitochondrial transcript percentage (`pct_counts_mt`) $< 15\%$
- hemoglobin transcript percentage (`pct_counts_hb`) $< 5\%$

Doublet detection

Following basic QC filtering, potential doublets were identified independently within each sample using Scrublet. Running Scrublet separately for each sample avoided estimating a single doublet-score distribution across biologically and technically distinct libraries.

For every cell, the Scrublet doublet score and predicted doublet status were stored in the AnnData metadata. Cells classified by Scrublet as predicted doublets were removed prior to downstream analysis. Doublet-score distributions and differences in total counts and detected genes between predicted singlets and predicted doublets were visualized as diagnostic quality-control plots.

After doublet removal, genes detected in fewer than five cells were excluded using `sc.pp.filter_genes(min_cells=5)` to remove extremely sparse features.

Quality-control metrics were recalculated following filtering and doublet removal. Sample-level and donor-type-level QC summaries, final cell counts, and a filtering audit table recording cell and gene counts at each processing stage were saved. The resulting QC-filtered AnnData object was saved as:

`merged_qc_filtered.h5ad`

and used as the input for normalization and feature selection.

2.2 Normalization and Highly Variable Gene Selection

The original count matrix was preserved in `adata.layers["counts"]` before expression transformation.

To account for differences in sequencing depth between cells, total-count normalization was performed using `sc.pp.normalize_total()` with a target sum of 10,000 counts per cell. Normalized expression values were then log-transformed using:

$x' = \log(1+x)$

via `sc.pp.log1p()`.

Before highly variable gene subsetting, the complete normalized and log-transformed expression matrix was stored in `adata.raw`. This preserved expression values for genes that might not be selected as highly variable but were required later for canonical lineage-marker visualization and interferon-response scoring.

Highly variable genes were identified using `sc.pp.highly_variable_genes()`, selecting the top 2,500 genes and specifying:

`batch_key="sample"`

to account for sample-specific variation during feature selection.

The `AnnData` object was then restricted to the selected 2,500 HVGs before computationally intensive regression and scaling.

Variation associated with sequencing depth and mitochondrial content was regressed out using:

- `total_counts`
- `pct_counts_mt`

with `sc.pp.regress_out()`.

The resulting expression matrix was subsequently standardized using `sc.pp.scale(max_value=10)`. This transformation centers and scales gene expression while clipping extreme scaled values at an absolute value of 10 to reduce the influence of outliers during principal component analysis.

The final processed HVG dataset was saved as:

`normalized_hvg_regressed_scaled.h5ad`

2.3 Dimensionality Reduction, Clustering, and Immune-Lineage Marker Assessment

Principal Component Analysis was performed on the scaled 2,500-HVG expression matrix using `sc.tl.pca()`. Fifty principal components were calculated using the ARPACK solver with a fixed random seed to improve reproducibility.

The proportion of variance explained by each principal component and cumulative variance contribution were recorded. Based on the PCA representation, a cell-neighborhood graph was constructed using the first 30 principal components and 20 nearest neighbors per cell with `sc.pp.neighbors()`.

Uniform Manifold Approximation and Projection was then applied using `sc.tl.umap()` to produce a two-dimensional visualization of the cell-neighborhood structure. Fixed random seeds were used for PCA, neighborhood construction, UMAP, and clustering to make repeated executions of the workflow more reproducible.

Leiden community detection was performed on the neighborhood graph at a resolution of 1.0 using `sc.tl.leiden()`. The resulting cluster assignments were stored in:

```
adata.obs["leiden_res_1.0"]
```

Cluster size and cluster composition by sample and donor type were exported for inspection.

Canonical marker genes were used to assess the major circulating immune lineages:

- **CD4+ T cells:** CD3D, CD4, IL7R, CCR7
- **CD8+ T cells:** CD3D, CD8A, CD8B, GZMB
- **Natural killer cells:** NKG7, GNLY, NCAM1
- **B cells:** MS4A1, CD19, CD79A
- **Monocytes:** CD14, FCGR3A, LYZ
- **Dendritic cells:** FCER1A, CST3

Marker availability was first evaluated within the 2,500-HVG matrix. Because several canonical lineage markers were not necessarily selected as HVGs, marker expression was additionally examined using the full normalized/log-transformed expression matrix stored in `adata.raw`.

Marker expression was visualized using cluster-level dot plots and UMAP feature plots. UMAPs colored by Leiden cluster, sample, and donor type were also examined to determine whether transcriptional structure was broadly shared across samples or disproportionately associated with individual donors.

The resulting clustered AnnData object was saved as:

```
clustering_res1.0.h5ad
```

2.4 Quantification of Interferon Response Capacity

Type I Interferon Response Capacity was quantified using a six-gene interferon-stimulated gene signature consisting of:

- BST2
- EIF2AK2 (PKR)
- ISG15
- MX1
- IFIT3
- IRF7

The availability of all six signature genes was checked against the complete normalized expression matrix stored in `adata.raw`.

A per-cell IRC module score was calculated using Scanpy's `sc.tl.score_genes()` with `use_raw=True`, ensuring that scoring was performed using the full normalized/log-transformed gene set rather than only the selected HVGs.

Scanpy's `score_genes()` method calculates a relative module-expression score rather than a simple arithmetic mean. The average expression of the target gene set is compared with the average expression of control genes selected from expression-matched gene bins. This approach reduces the influence of background gene-expression levels when estimating signature activity.

The resulting score was stored as:

```
adata.obs["IRC_score"]
```

Overall descriptive statistics and sample-level IRC-score summaries were exported. IRC scores were visualized on the existing UMAP embedding and compared across samples and donor types using violin plots.

The IRC-scored AnnData object was saved as:

```
ifn_irc_scored.h5ad
```

2.5 Cell-Level IRC Stratification and Differential Expression

To examine transcriptional differences associated with high and low IRC states, the AnnData object was restricted to cells originating from cancer patients:

```
donor_type == "Patient"
```

The median IRC score across all patient-derived cells was calculated and used as the cell-level stratification threshold.

Patient cells were classified as:

- **IRC-High:** IRC score greater than or equal to the median patient-cell IRC score
- **IRC-Low:** IRC score below the median patient-cell IRC score

This procedure represents **cell-level stratification rather than patient-level stratification**. Individual patients were not assigned a single IRC-High or IRC-Low classification.

Differential expression between IRC-High and IRC-Low patient cells was performed using the Wilcoxon rank-sum test implemented in `sc.tl.rank_genes_groups()`, with IRC-Low specified as the reference group and expression values taken from `adata.raw`.

The IRC-High differential-expression results were extracted using `sc.get.rank_genes_groups_df()`. Adjusted p-values reported by Scanpy were used to account for multiple testing.

Because multiple cells originate from the same biological sample, the resulting analysis should be interpreted as an **exploratory cell-level differential-expression comparison**, rather than as a patient-level statistical comparison based on independent biological replicates.

The complete differential-expression table was saved as:

`IRC_High_vs_Low_DE.csv`

and the patient-cell AnnData object containing IRC group assignments and DE results was saved as:

`patient_irc_DE.h5ad`

2.6 Over-Representation Pathway Enrichment Analysis

Differentially expressed genes were first classified as significant using the following criteria:

- adjusted p-value < 0.05
- absolute log2 fold change > 0.5

For pathway enrichment, genes specifically upregulated in IRC-High cells were selected using:

- log2 fold change > 0.5
- adjusted p-value < 0.05

Up to the first 100 genes satisfying these criteria in the ranked differential-expression table were used as the input gene list for over-representation analysis.

Pathway enrichment was performed using GSEAPy's `gp.enrichr()` interface. Three Enrichr gene-set libraries were queried:

1. Reactome 2022
2. KEGG 2021 Human
3. GO Biological Process 2023

Because Enrichr is accessed as an online service through GSEAPy, an active internet connection was required for this stage of the workflow.

The complete enrichment output was saved as:

`pathway_enrichment_results.csv`

The fifteen pathways with the smallest adjusted p-values were additionally extracted for visualization. Pathway significance was displayed as:

$-\log_{10}(\text{adjusted p-value})$

in a horizontal bar plot.

Differential-expression results were also visualized using volcano plots with significance thresholds at:

- $\log_2 \text{ fold change} = \pm 0.5$
- $\text{adjusted p-value} = 0.05$

A second volcano plot labeled the ten most statistically significant genes among those passing the predefined significance thresholds.

Because the pathway analysis was based on genes identified from the exploratory cell-level IRC-High versus IRC-Low comparison, the enrichment results were likewise interpreted as exploratory evidence of biological programs associated with high IRC states rather than as independent patient-level pathway differences.

3. Results

3.1 Dataset Overview and Quality Control

The initial merged single-cell RNA-sequencing dataset contained 101,020 cells and 36,601 genes derived from two healthy donors (HD1 and HD2) and eight cancer patients (P1–P8).

Quality-control metrics were evaluated before filtering to assess the distributions of detected genes, total transcript counts, mitochondrial transcript percentage, ribosomal transcript percentage, and hemoglobin transcript percentage across individual samples and donor groups.

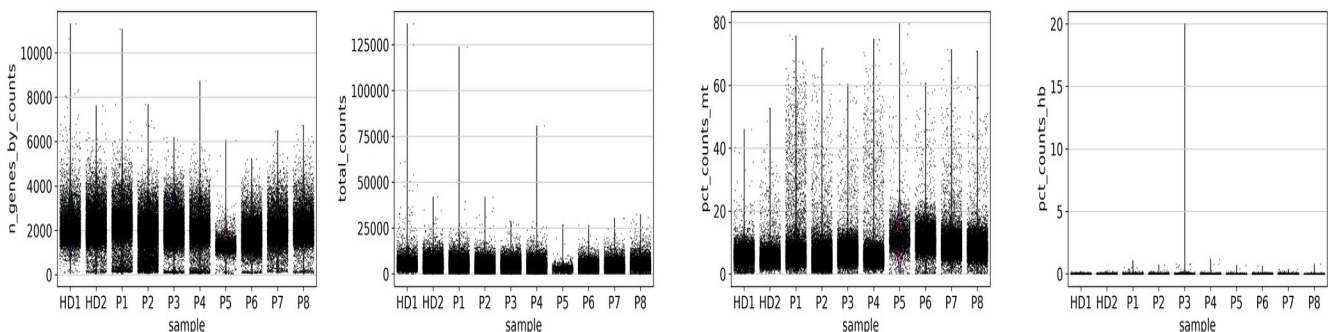


Figure 1. Quality-control metrics before filtering across individual samples.

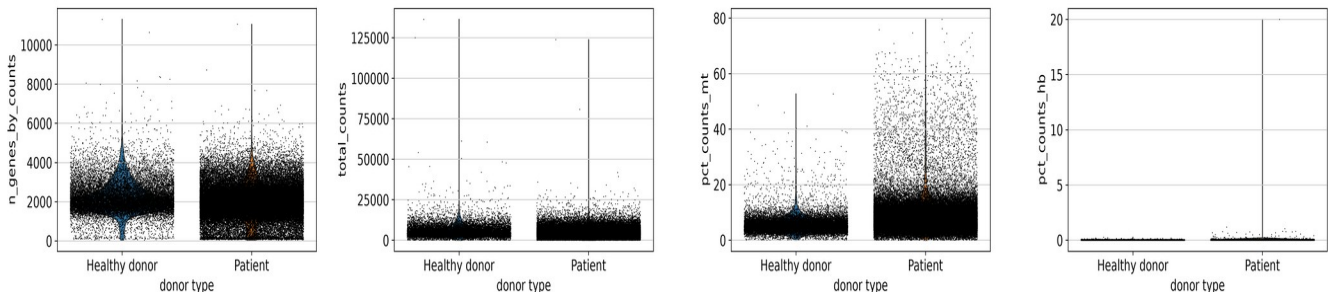


Figure 2. Quality-control metrics before filtering according to donor type.

Sample-level quality-control statistics before filtering were summarized numerically.

sample	n_genes_by_counts	total_counts	pct_counts_mt	pct_counts_hb
HD1	2033	4412	5.821004	0
HD2	2068	4194	4.6366415	0
P1	2245	5049	6.135209	0
P2	1909	4022	5.0111356	0
P3	1996.5	4317	6.621499	0
P4	1984	4513	5.2726088	0

P5	1298	2545	11.416914	0
P6	1678	3454.5	10.859177	0
P7	1960.5	3965.5	8.808832	0
P8	1970	4327	8.525476	0

Table 1. Sample-level quality-control summary before filtering.

Quality-control statistics were also summarized after grouping cells according to donor type to compare the overall pre-filtering distributions between healthy-donor and patient-derived cells.

donor_type	n_genes_by_counts	total_counts	pct_counts_mt	pct_counts_hb
Healthy donor	2055	4293	5.0495515	0
Patient	1946	4189	7.076102	0

Table 2. Quality-control summary before filtering according to donor type.

Cells were filtered according to the predefined QC thresholds of at least 300 detected genes per cell, less than 15% mitochondrial transcripts, and less than 5% hemoglobin transcripts.

Following basic QC filtering, potential doublets were assessed independently within each sample using Scrublet. Scrublet assigned a doublet score and predicted doublet status to each cell.

Predicted doublets generally exhibited distinct distributions of detected genes and total transcript counts compared with predicted singlets.

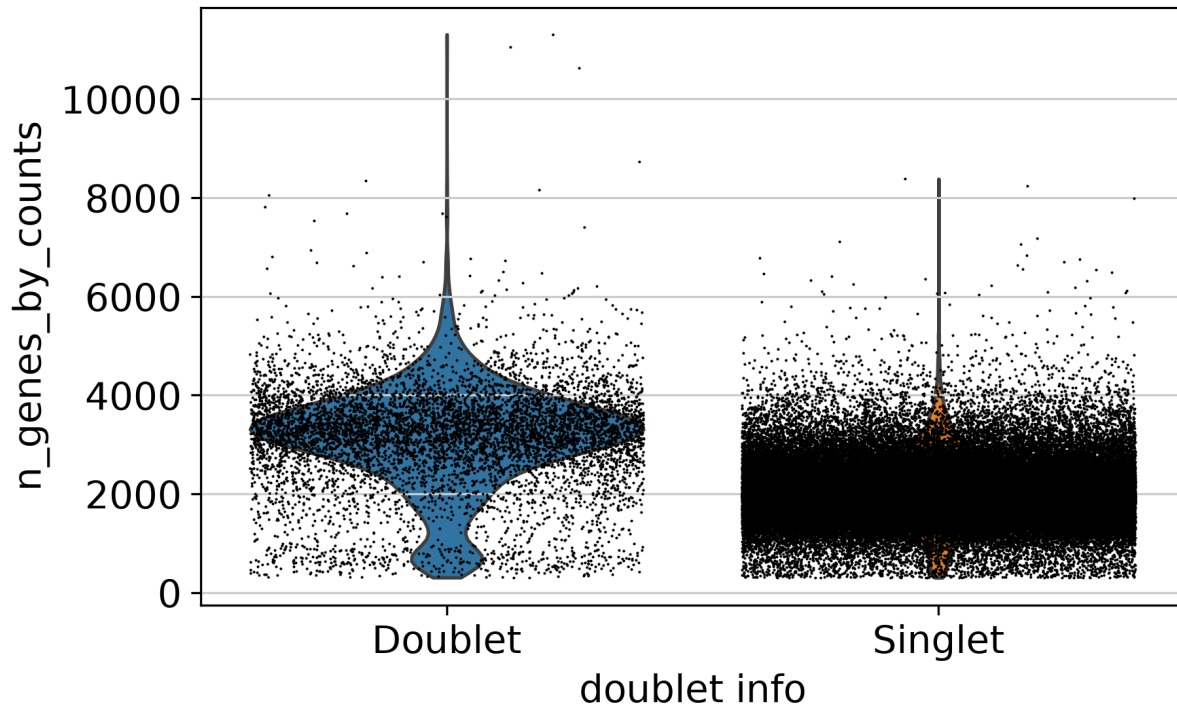


Figure 3. Distribution of detected genes in Scrublet-predicted singlets and doublets.

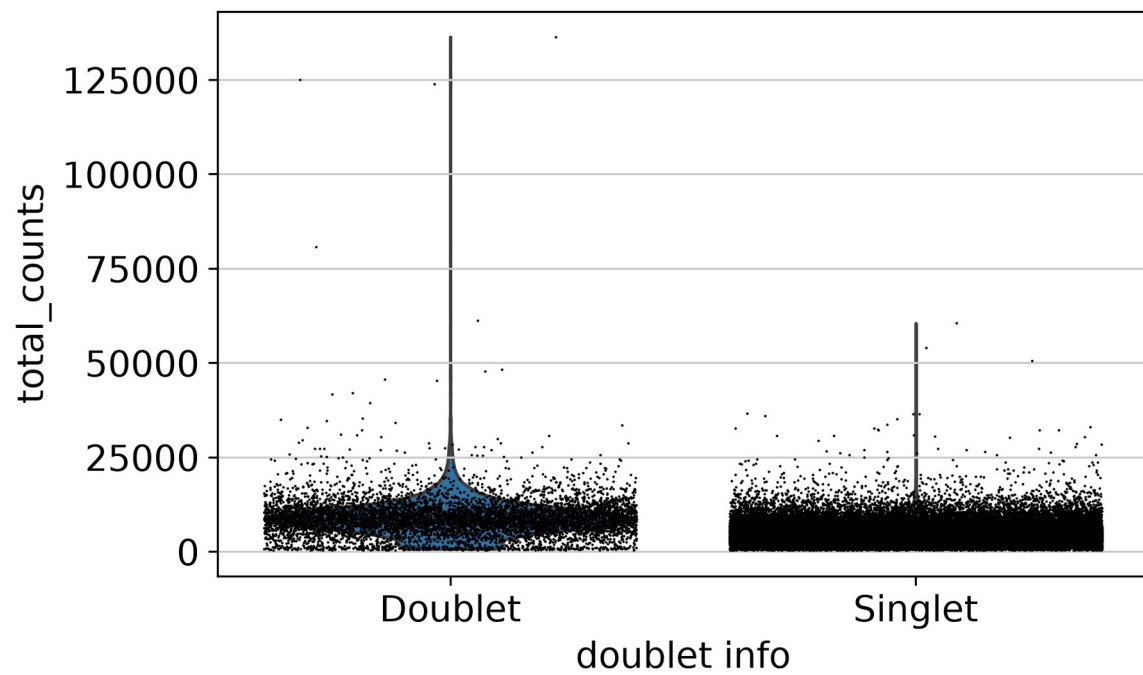


Figure 4. Distribution of total transcript counts in Scrublet-predicted singlets and doublets.

The distributions of Scrublet doublet scores were also evaluated across the ten individual samples.

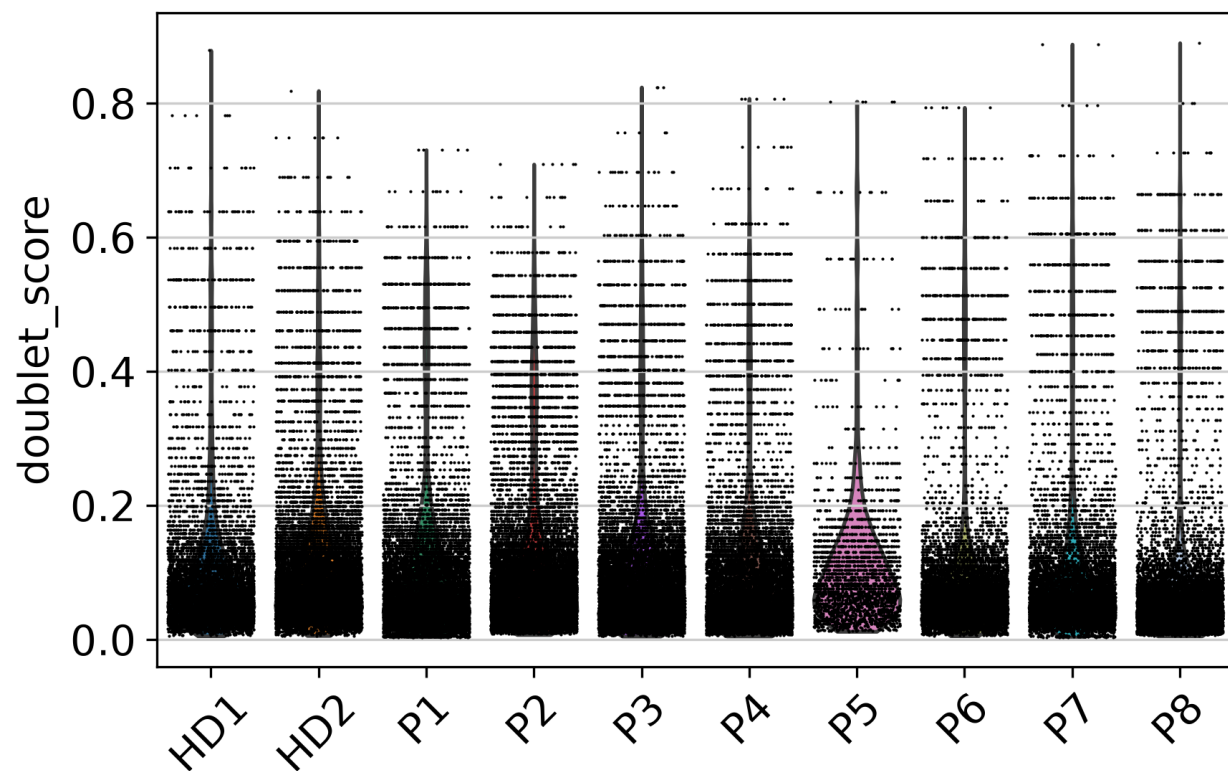


Figure 5. Scrublet doublet-score distributions across samples.

The number and proportion of Scrublet-predicted doublets in each sample were recorded prior to their removal.

sample	cells	predicted_doublets	median_doublet_score	doublet_rate_percent
HD1	7482	361	0.0672727272727273	4.82491312483293
HD2	11694	834	0.0762052877138414	7.13186249358646
P1	9718	810	0.0653029110936271	8.33504836386088
P2	13774	1482	0.0715268225584594	10.7594017714535
P3	12590	949	0.0657439446366782	7.53772835583797
P4	10194	779	0.0581395348837209	7.64175004904846
P5	2357	80	0.0836363636363636	3.39414509970301
P6	8644	553	0.0570758405003909	6.39750115687182
P7	8850	585	0.0577367205542725	6.61016949152542
P8	9210	632	0.0486787204450626	6.86210640608035

Table 3. Number and proportion of Scrublet-predicted doublets by sample.

Following removal of predicted doublets, genes detected in fewer than five cells were excluded. QC metrics were recalculated after filtering to evaluate the final cell population retained for downstream analysis.

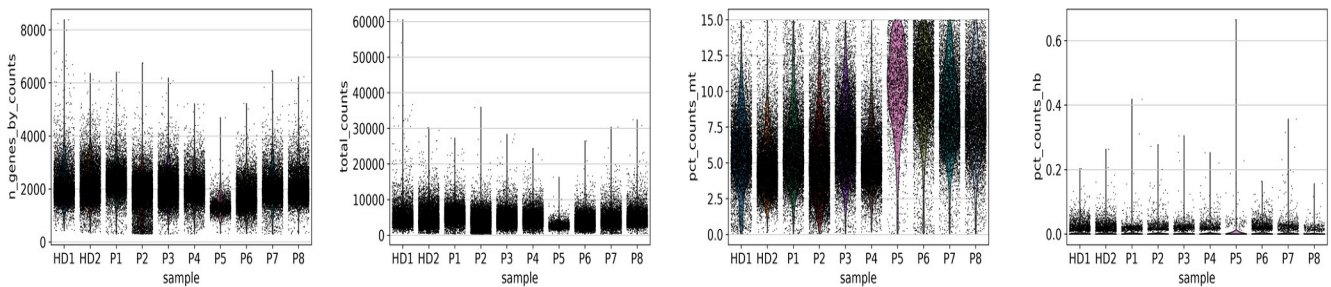


Figure 6. Quality-control metrics after filtering and doublet removal across individual samples.

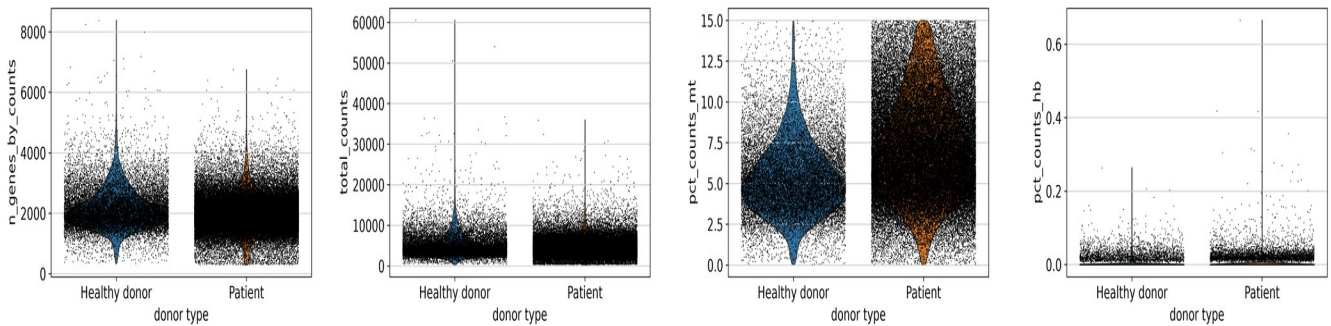


Figure 7. Quality-control metrics after filtering and doublet removal according to donor type.

The number of cells and genes retained at each major preprocessing stage was summarized in the filtering audit table.

stage	cells	genes
Raw merged data	101020	36601
After basic QC	94513	36601
After Scrublet doublet removal	87448	36601
After gene filtering	87448	30744

Table 4. Cell and gene retention across the quality-control workflow.

The final QC-filtered dataset was saved as `merged_qc_filtered.h5ad` and used as the input for subsequent normalization and feature-selection steps.

3.2 Normalization and Highly Variable Gene Selection

Following quality control, expression values were normalized to 10,000 counts per cell and log-transformed. The complete normalized expression matrix was retained in `adata.raw`.

Highly variable gene selection identified 2,500 genes for downstream dimensionality reduction and clustering while accounting for sample identity during feature selection.

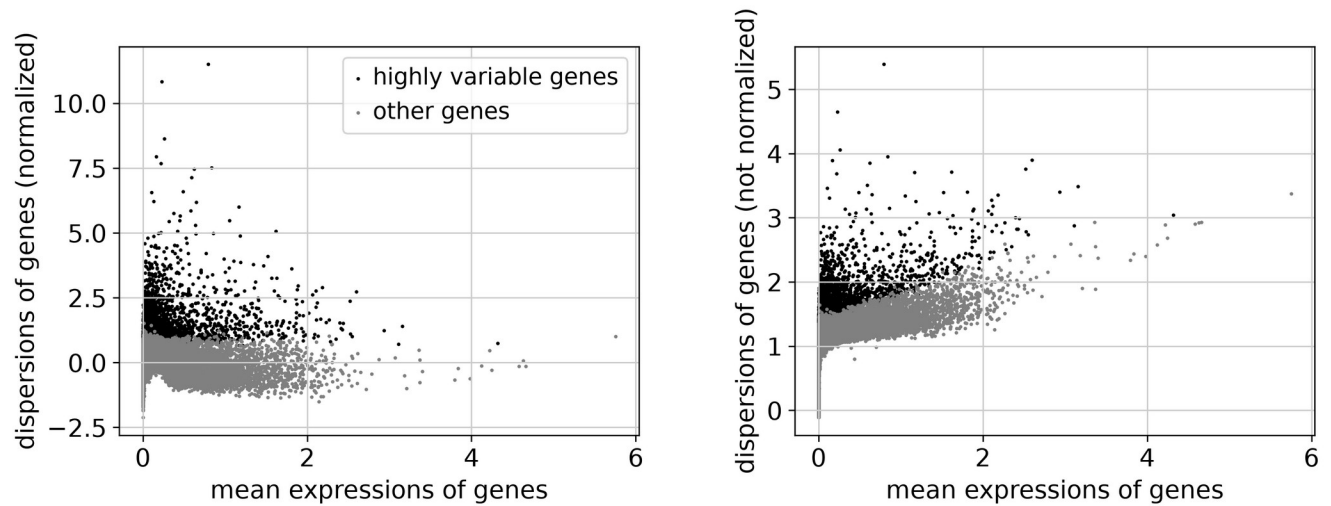


Figure 8. Highly variable gene selection following normalization and log transformation.

The number of genes evaluated and selected as highly variable was recorded.

metric	value
total_genes	36601
selected_hvgs	2500
requested_hvgs	2500

Table 5. Summary of highly variable gene selection.

After restricting the dataset to the selected HVGs, variation associated with total transcript counts and mitochondrial transcript percentage was regressed out, and gene expression values were scaled before dimensionality reduction.

3.3 Principal Component Analysis and Leiden Clustering

Principal Component Analysis was performed on the scaled 2,500-HVG expression matrix, generating 50 principal components.

The amount of variance explained by each component decreased progressively across the first 50 PCs.

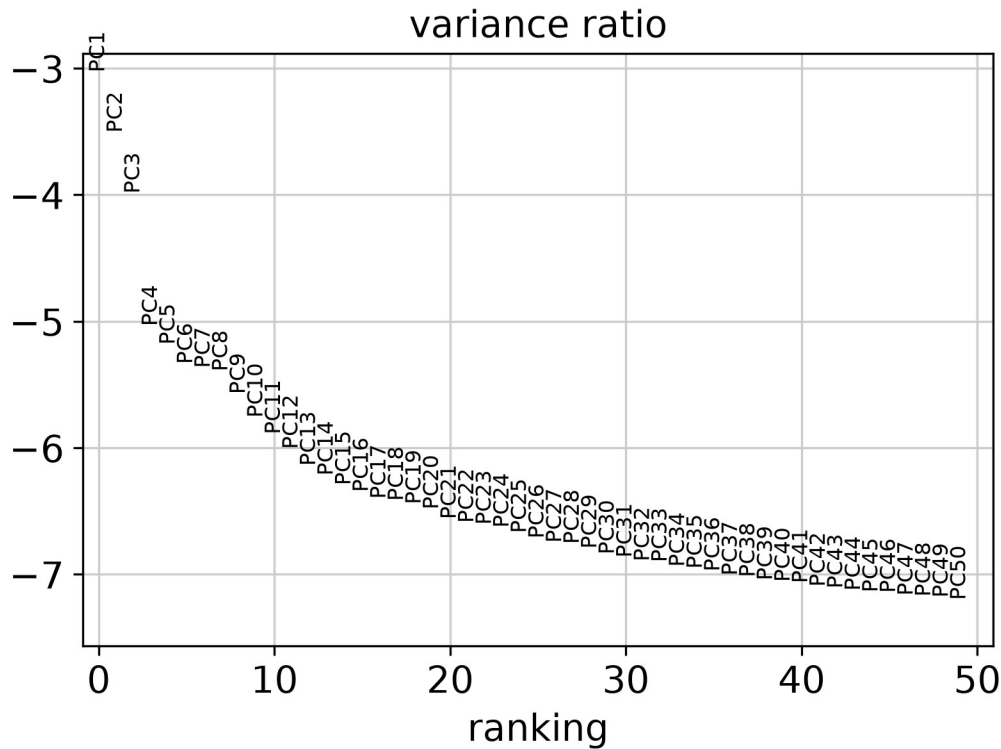


Figure 9. Variance explained by the first 50 principal components.

A neighborhood graph was constructed using the first 30 PCs and 20 nearest neighbors per cell. UMAP was subsequently used to visualize the transcriptional structure of the dataset.

Leiden clustering at a resolution of 1.0 identified **30 transcriptionally distinct clusters**.

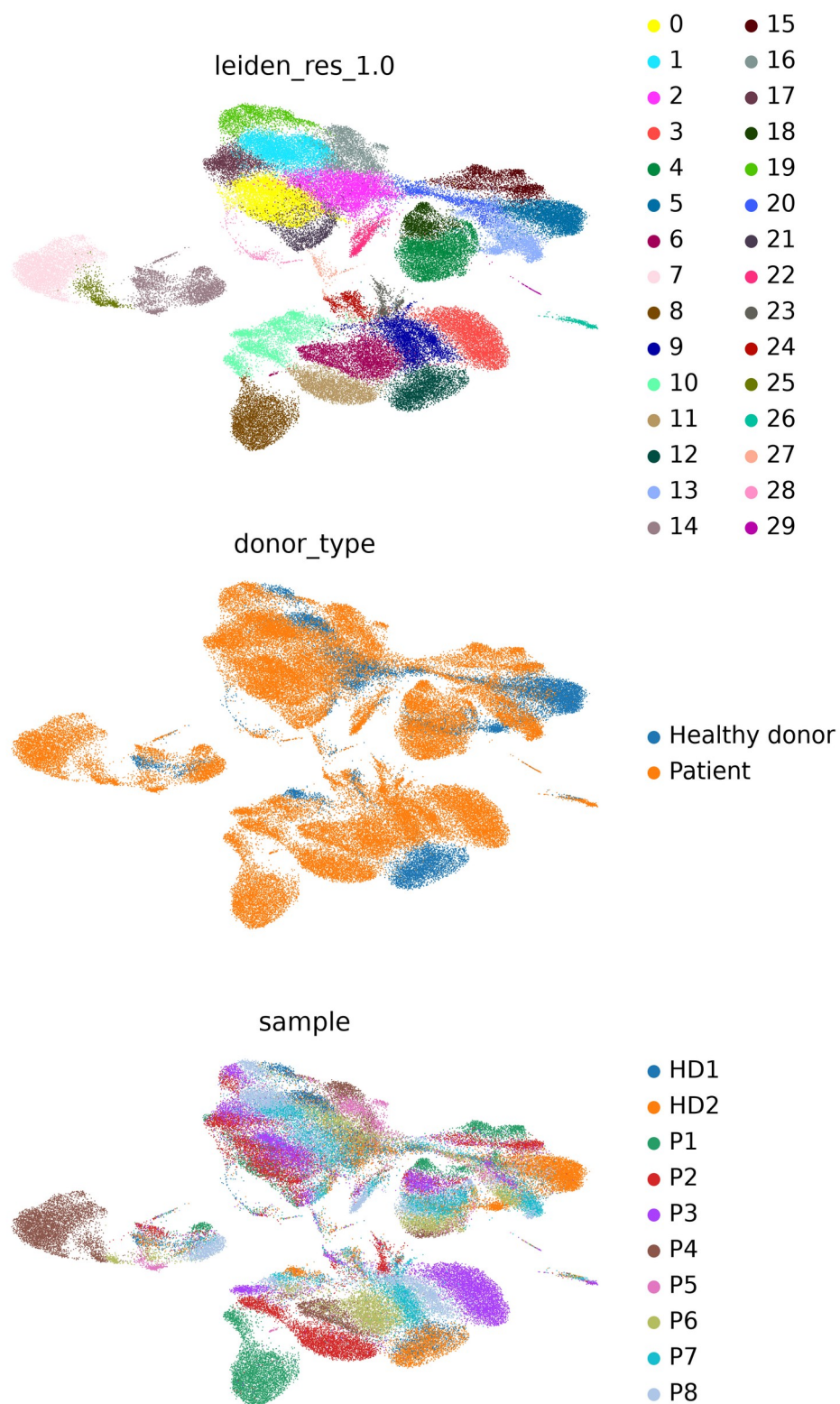


Figure 10. UMAP representation of PBMCs colored by Leiden cluster, donor type, and sample identity.

The number and relative proportion of cells assigned to each Leiden cluster were quantified.

leiden_res_1.0	cells	fraction_of_cells
0	5826	0.0666224499130912
1	5618	0.0642438935138597
2	5617	0.0642324581465557
3	5442	0.0622312688683561
4	5396	0.0617052419723722
5	4915	0.0562048302991492
6	4612	0.0527399140060379
7	4441	0.0507844661970543
8	3977	0.0454784557679993
9	3931	0.0449524288720154
10	3614	0.0413274174366481
11	3569	0.0408128259079682
12	3541	0.0404926356234562
13	3499	0.0400123501966883
14	3307	0.0378167596743207
15	2830	0.0323620894703138
16	2777	0.0317560150032019
17	2361	0.0269989022047388
18	2000	0.0228707346079956
19	1986	0.0227106394657396
20	1932	0.0220931296313238
21	1783	0.0203892599030281
22	966	0.0110465648156619
23	892	0.010200347635166
24	800	0.00914829384319824
25	734	0.00839355960113439
26	355	0.00405955539291922
27	333	0.00380797731223127
28	319	0.0036478821699753
29	75	0.000857652547799835

Table 6. Cell counts and relative abundance of Leiden clusters.

Cluster composition was also examined across individual samples.

leiden_res_1.0	HD1	HD2	P1	P2	P3	P4	P5	P6	P7	P8
0	17	17	1187	2588	1844	7	0	3	86	77
1	1479	190	13	104	240	45	12	392	1396	1747
2	1557	1094	0	20	62	296	202	692	1163	531
3	1	2	5	14	5252	0	0	3	28	137
4	317	710	19	72	99	632	145	1460	1315	627
5	284	4303	3	7	17	3	9	28	81	180
6	5	7	1	24	8	1577	151	2711	31	97
7	0	0	0	1	1	4437	0	1	0	1
8	2	0	3944	13	12	0	0	0	5	1
9	37	25	28	25	70	5	1	210	1727	1803
10	123	364	510	988	647	113	15	262	295	297

11	7	3	19	3396	45	26	2	24	33	14
12	1406	2035	0	5	4	12	1	7	63	8
13	202	811	352	19	327	19	12	672	798	287
14	397	277	333	387	82	32	303	187	207	1102
15	7	199	1156	1167	210	5	6	4	19	57
16	11	5	0	3	2	1243	951	537	8	17
17	1	2	277	1289	719	1	0	0	28	44
18	8	48	555	497	810	1	0	3	44	34
19	391	32	28	323	404	79	59	77	52	541
20	520	288	1	7	11	141	339	304	201	120
21	92	102	239	410	278	126	41	129	241	125
22	44	22	24	74	74	4	2	7	147	568
23	17	24	44	618	36	18	11	28	77	19
24	65	146	42	85	175	21	3	82	96	85
25	0	0	0	3	0	559	0	172	0	0
26	32	48	20	62	90	0	3	36	46	18
27	20	56	42	27	72	8	2	40	56	10
28	75	37	64	52	27	3	5	11	17	28
29	4	13	2	12	23	2	2	9	5	3

Table 7. Distribution of individual samples across Leiden clusters.

Cluster composition was also summarized according to donor type to compare the contribution of healthy-donor and patient-derived cells within each cluster.

leiden_res_1.0	Healthy donor	Patient	leiden_res_1.0	Healthy donor	Patient
0	34	5792	15	206	2624
1	1669	3949	16	16	2761
2	2651	2966	17	3	2358
3	3	5439	18	56	1944
4	1027	4369	19	423	1563
5	4587	328	20	808	1124
6	12	4600	21	194	1589
7	0	4441	22	66	900
8	2	3975	23	41	851
9	62	3869	24	211	589
10	487	3127	25	0	734
11	10	3559	26	80	275
12	3441	100	27	76	257
13	1013	2486	28	112	207
14	674	2633	29	17	58

Table 8. Distribution of healthy-donor and patient-derived cells across Leiden clusters.

3.4 Canonical Immune-Lineage Marker Expression

Canonical marker expression was evaluated across Leiden clusters to assess the presence of the major circulating immune-cell populations expected within the PBMC dataset.

The marker panel included genes associated with CD4⁺ T cells, CD8⁺ T cells, natural killer cells, B cells, monocytes, and dendritic cells.

Marker expression was first examined among genes present in the selected HVG matrix.

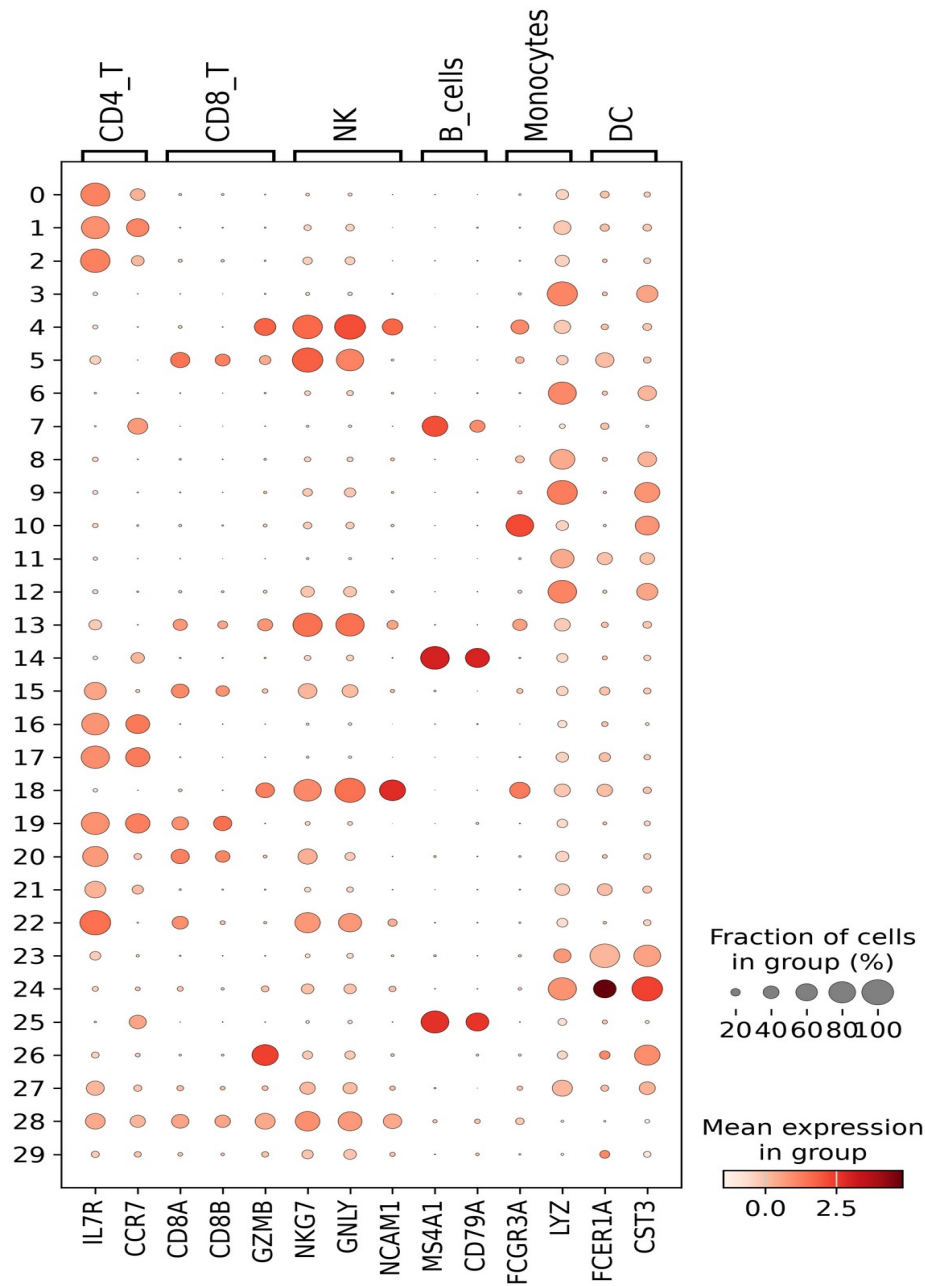


Figure 11. Canonical immune-lineage marker expression across Leiden clusters using the HVG matrix.

Selected immune markers were also visualized directly on the UMAP embedding.

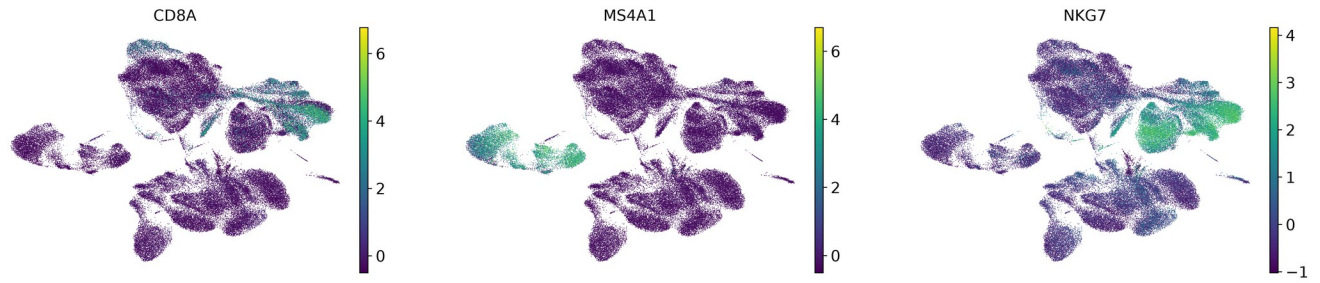


Figure 12. UMAP feature plots of canonical immune-lineage markers present in the HVG matrix.

Because several canonical markers were not necessarily retained among the 2,500 HVGs, marker expression was additionally assessed using the complete normalized expression matrix stored in `adata.raw`.

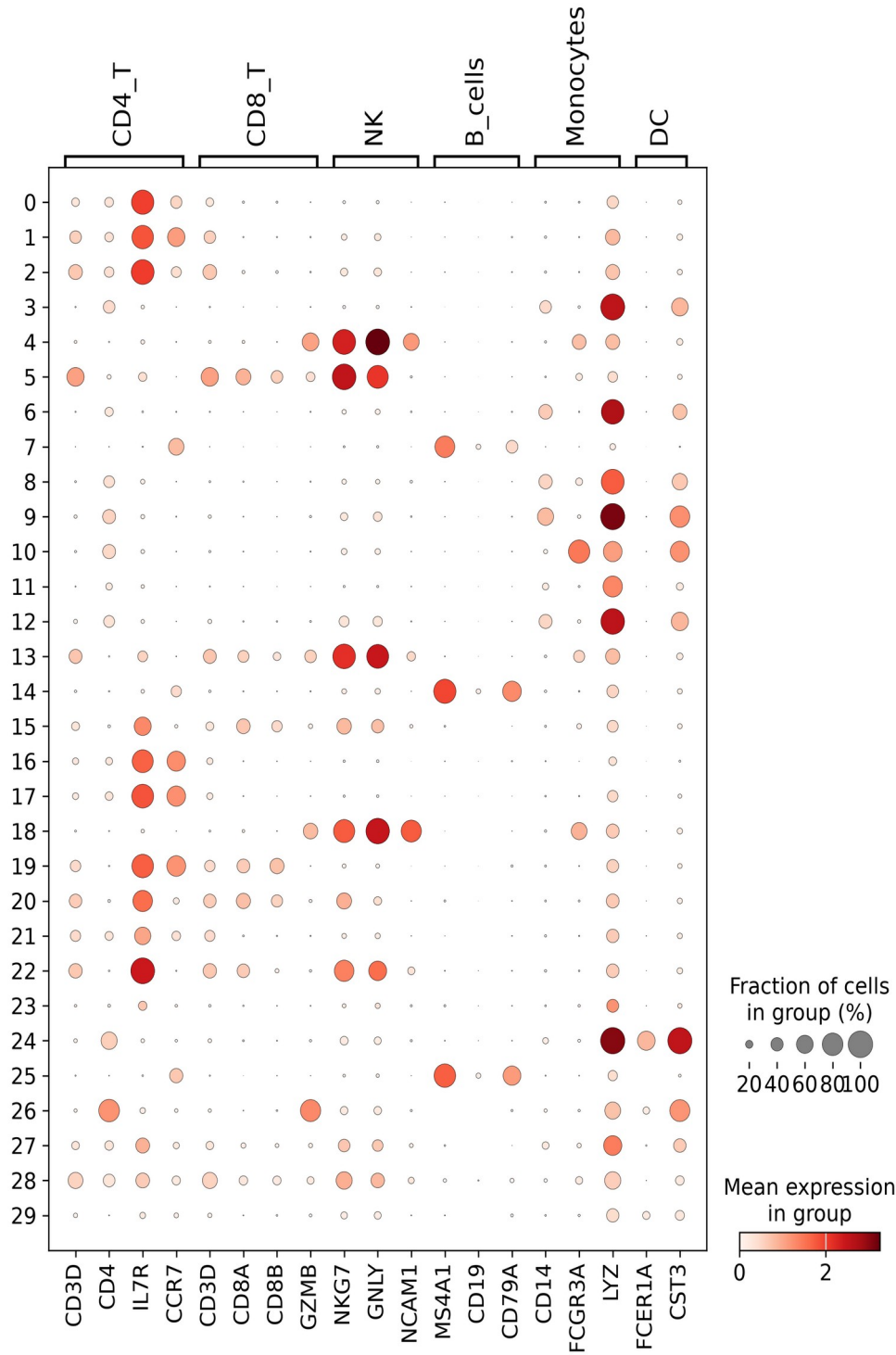


Figure 13. Canonical immune-lineage marker expression across Leiden clusters using the full normalized expression matrix.

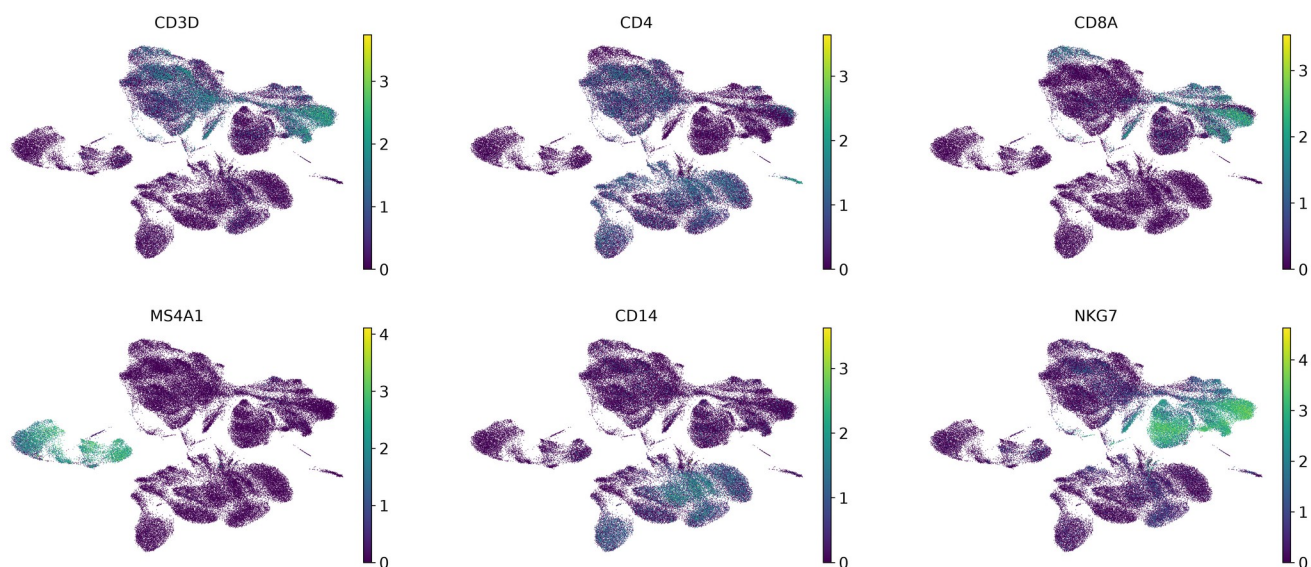


Figure 14. UMAP feature plots of canonical immune markers using the full normalized expression matrix.

The observed marker-expression patterns were used to evaluate the distribution of the major PBMC lineages across the identified Leiden clusters.

3.5 Interferon Response Capacity Across Cells and Samples

A cell-level Interferon Response Capacity score was calculated using the six-gene signature consisting of BST2, EIF2AK2, ISG15, MX1, IFIT3, and IRF7.

The availability of the six signature genes in the complete normalized expression matrix was verified before score calculation.

gene	available_in_raw
BST2	True
EIF2AK2	True
ISG15	True
MX1	True
IFIT3	True
IRF7	True

Table 9. Availability of IRC-signature genes in the normalized expression matrix.

IRC scores varied across individual cells within the PBMC dataset.

The spatial distribution of IRC activity across the UMAP embedding was visualized alongside donor type and Leiden cluster identity.

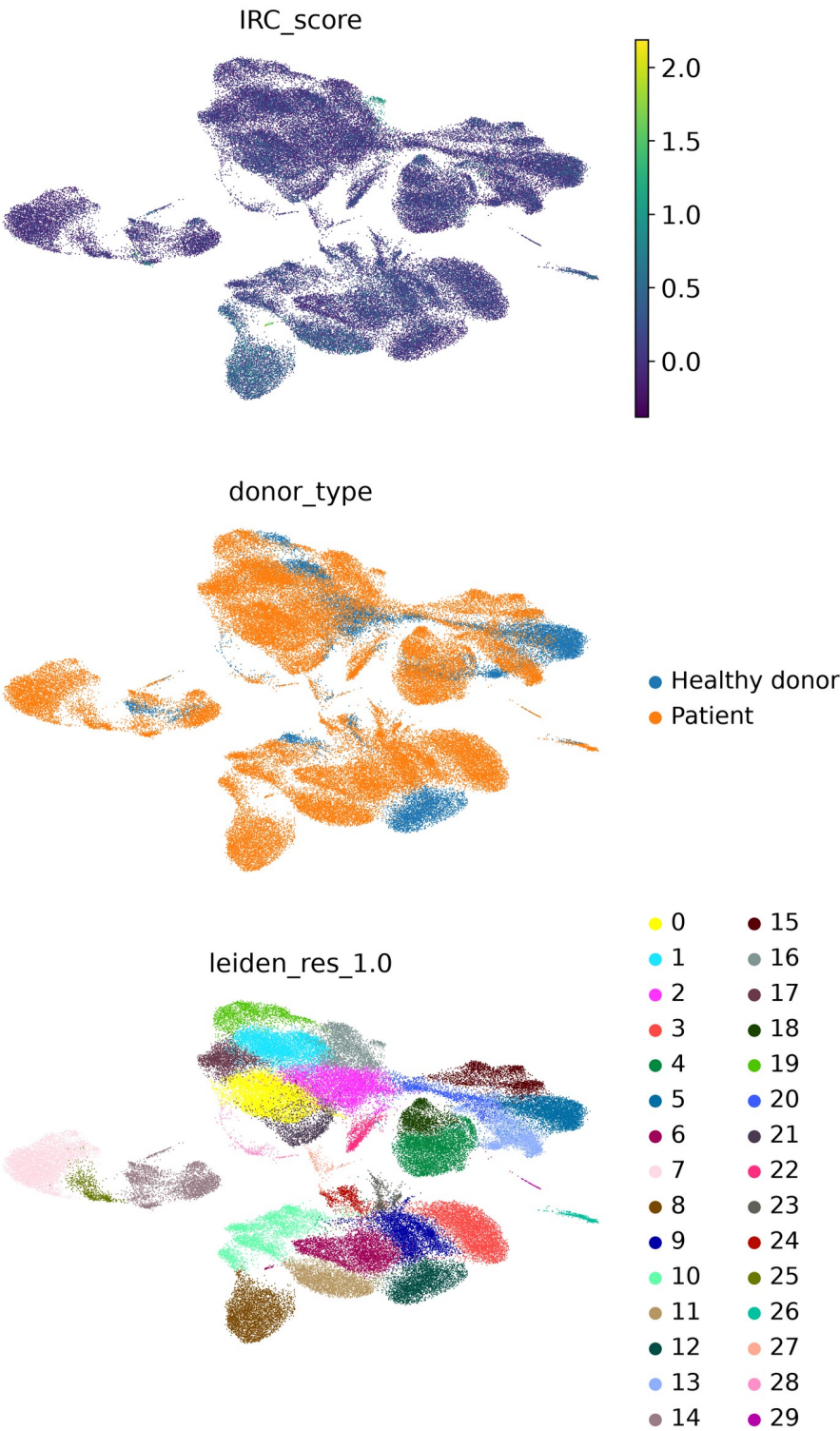


Figure 15. UMAP visualization of cell-level IRC score, donor type, and Leiden cluster identity.

Variation in IRC-score distributions was also observed across individual samples.

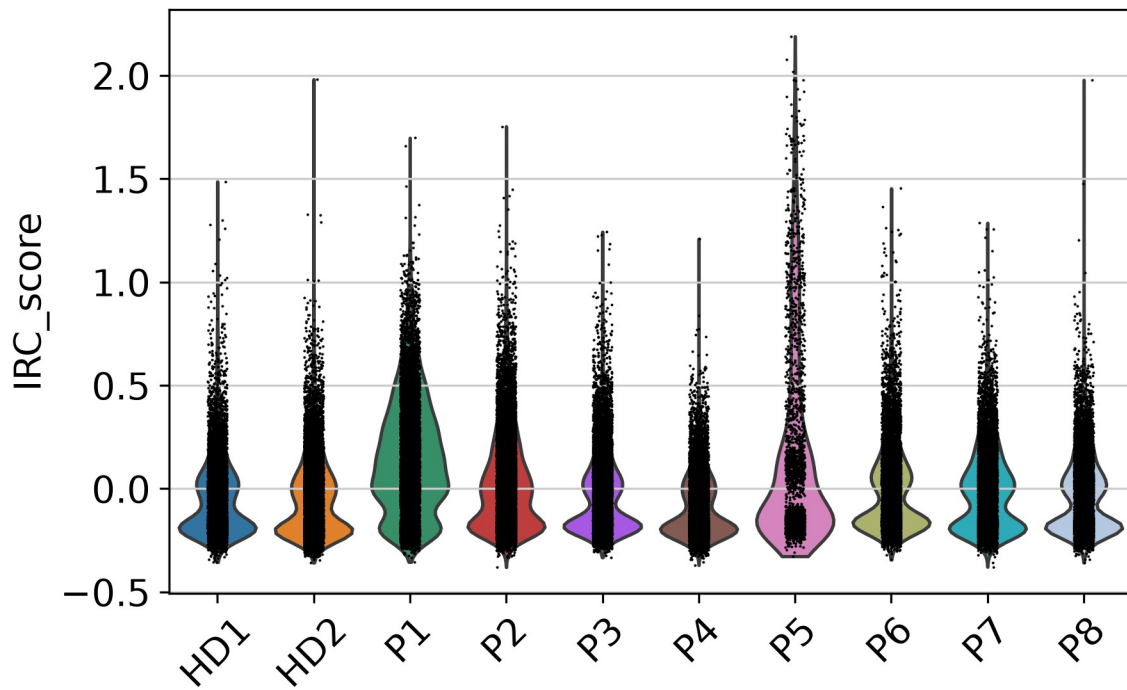


Figure 16. Distribution of cell-level IRC scores across individual samples.

IRC-score distributions were additionally compared between healthy-donor and patient-derived cells.

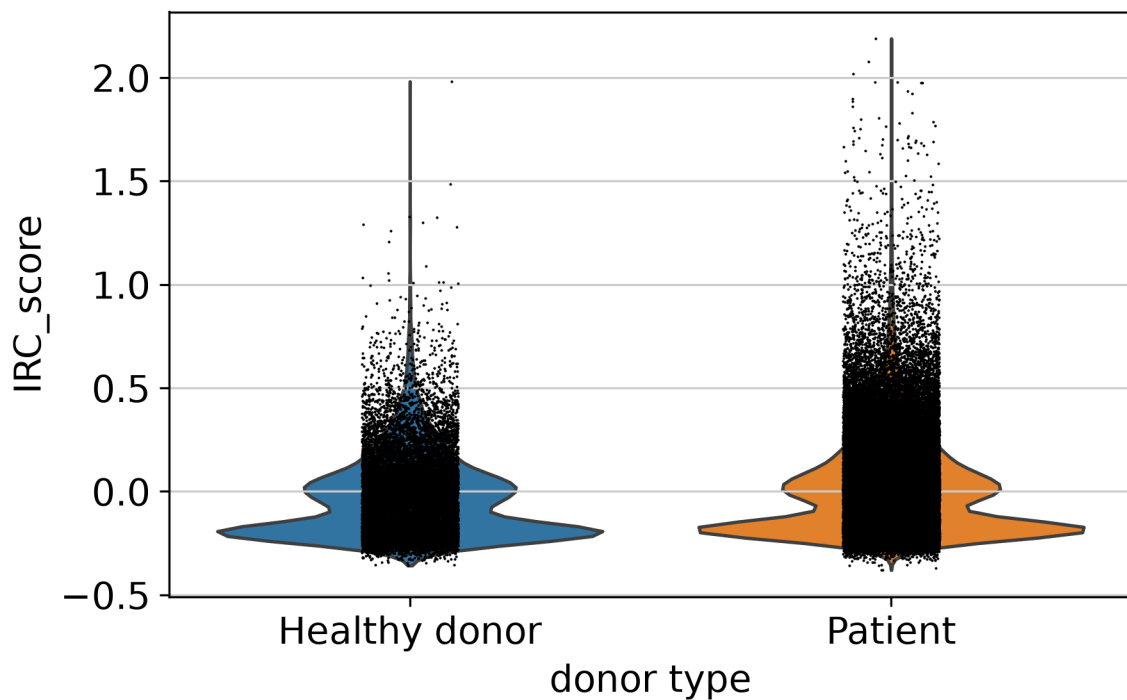


Figure 17. Distribution of cell-level IRC scores according to donor type.

Sample-level descriptive statistics for IRC scores were calculated to quantify differences in central tendency and dispersion across samples.

sample	cells	mean	median	std	minimum	maximum
HD1	7121	-0.0349326931465455	-0.0785609730084737	0.197871939805721	-0.356183815002441	1.48614833196004
HD2	10860	-0.0356045519451081	-0.0676450498898824	0.197644222101695	-0.358655281066895	1.98080629984538
P1	8908	0.162016661040682	0.126783762772878	0.281581506121571	-0.355252265930176	1.69744697570801
P2	12292	0.0345735755422645	-0.0119488469759623	0.237262406924032	-0.380510749816895	1.75310003598531
P3	11641	-0.0247353249858432	-0.0638472326596578	0.19395969701992	-0.33297046661377	1.24253933588664
P4	9415	-0.0846318913377224	-0.138890581130981	0.154883568996356	-0.370544509887695	1.20838283538818
P5	2277	0.16956359552667	-0.0176821796099345	0.481548529682089	-0.327906494140625	2.18870644569397
P6	8091	0.0060170592509521	-0.0354124855995178	0.21558351629669	-0.344169235229492	1.45298433939616
P7	8265	-0.000367228882498836	-0.020020588239034	0.204670418565694	-0.378871650695801	1.28568935394287
P8	8578	-0.021469769118558	-0.0416214791933695	0.191267839478799	-0.358041877746582	1.97770778338114

Table 10. Sample-level descriptive statistics for cell-level IRC scores.

3.6 Cell-Level IRC-High and IRC-Low Stratification

The analysis was restricted to cells derived from cancer patients. The median IRC score across all patient cells was used as the threshold for cell-level stratification.

Patient cells with IRC scores greater than or equal to the median were classified as IRC-High, whereas cells with scores below the median were classified as IRC-Low.

IRC_group	cells	fraction_of_patient_cells	median_cutoff
IRC_High	34734	0.500007197662199	-0.027669734954834
IRC_Low	34733	0.499992802337801	-0.027669734954834

Table 11. Number and proportion of patient cells classified as IRC-High and IRC-Low.

This grouping represented a cell-level transcriptional classification rather than assignment of entire patients to IRC-High or IRC-Low categories.

3.7 Differential Expression Between IRC-High and IRC-Low Patient Cells

Differential-expression analysis was performed between IRC-High and IRC-Low patient cells using the Wilcoxon rank-sum test.

Genes were ranked according to their differential-expression statistics, and adjusted p-values were used to account for multiple testing.

Genes with an adjusted p-value below 0.05 and an absolute log₂ fold change greater than 0.5 were considered significant for visualization and downstream enrichment analysis.

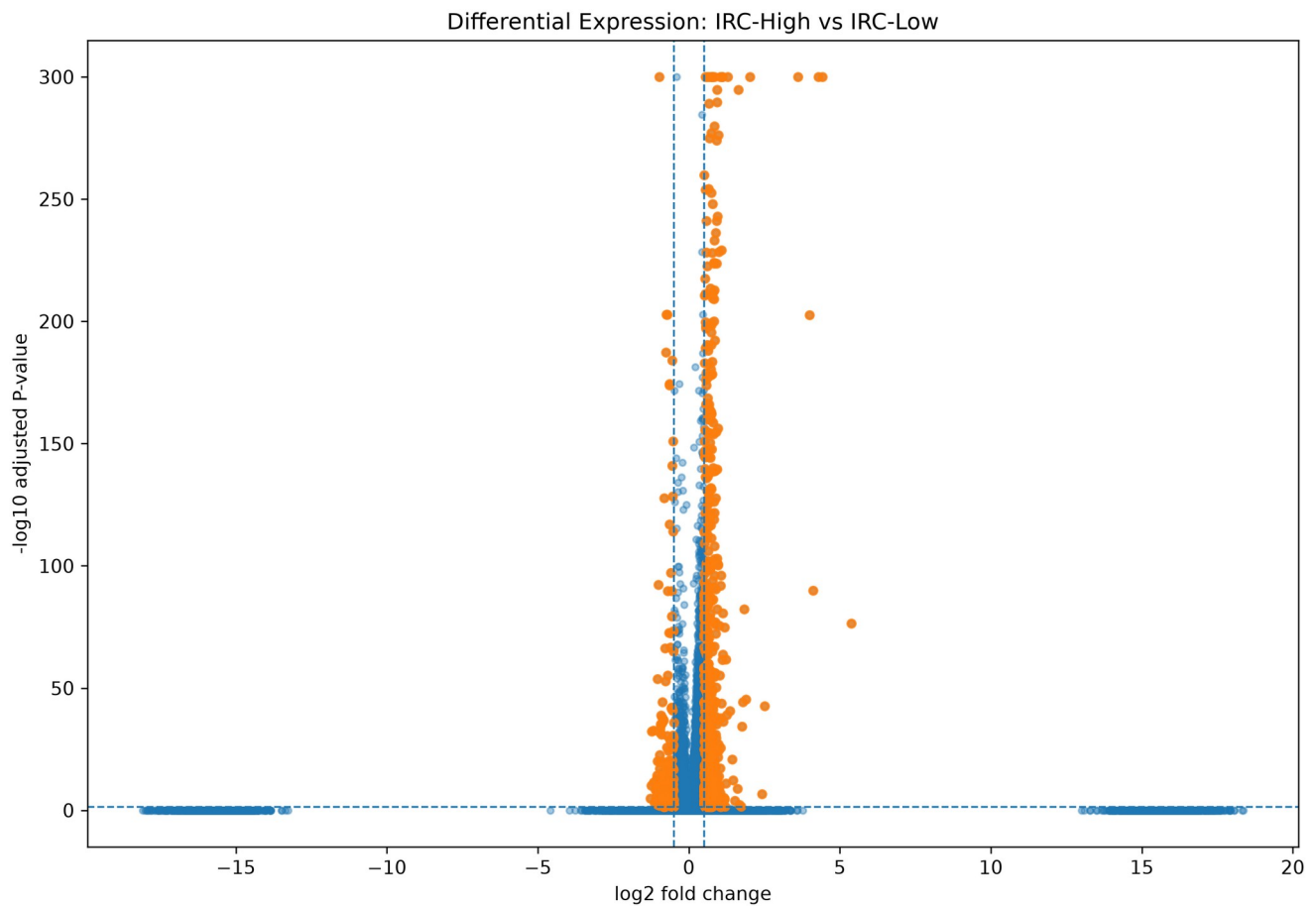


Figure 18. Volcano plot of differential expression between IRC-High and IRC-Low patient cells.

The ten most statistically significant genes among those passing the predefined significance criteria were labeled in a second volcano plot.

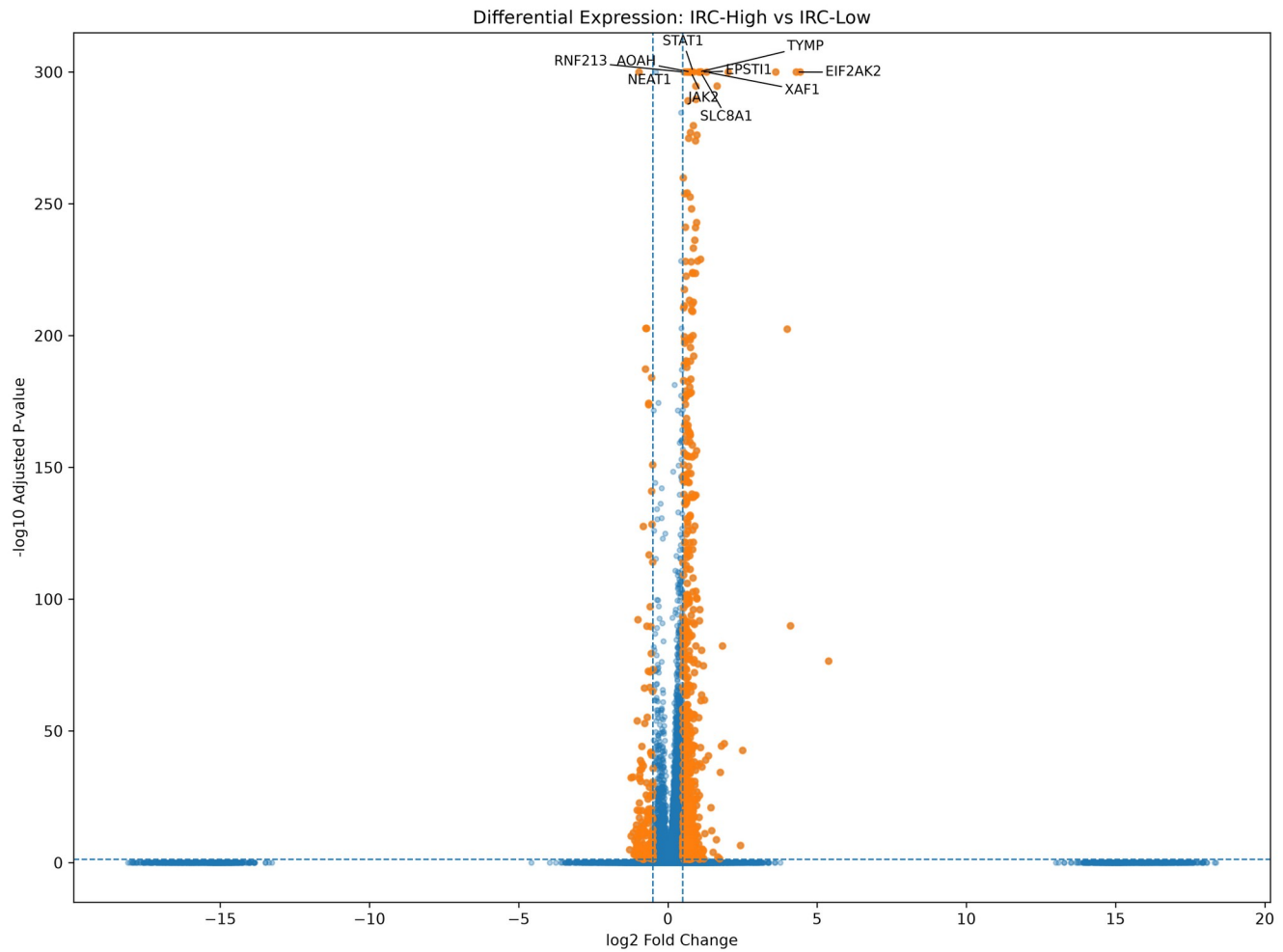


Figure 19. Labeled volcano plot highlighting prominent differentially expressed genes between IRC-High and IRC-Low patient cells.

A selected subset of the highest-ranking genes can be presented directly in the main text.

names	scores	logfoldchanges	pvals	pvals_adj
EIF2AK2	140.49612	4.2963433	0	0
MX1	81.22656	4.428481	0	0
BST2	60.476437	3.6067643	0	0
STAT1	41.792515	0.8495058	0	0
TYMP	40.79485	1.0309801	0	0
EPSTI1	40.77428	1.1166257	0	0
RNF213	40.08999	0.54803956	0	0
AOAH	39.043232	0.7763127	0	0
PARP14	38.989395	0.7752081	0	0
JAK2	38.946476	0.76822793	0	0

Table 12. Selected differentially expressed genes between IRC-High and IRC-Low patient cells (Top 10 which are labeled in Fig 19)

The complete differential-expression results were retained in `IRC_High_vs_Low_DE.csv`.

3.8 Pathway Enrichment of Genes Upregulated in IRC-High Cells

Genes significantly upregulated in IRC-High cells were selected using a log2 fold-change threshold greater than 0.5 and an adjusted p-value below 0.05.

Up to 100 genes passing these thresholds were used for over-representation analysis against Reactome 2022, KEGG 2021 Human, and GO Biological Process 2023 gene-set libraries.

The complete pathway-enrichment output was saved in `pathway_enrichment_results.csv`.

The 15 pathways with the smallest adjusted p-values were selected for visualization.

Gene_set	Term	P-value
GO_Biological_Process_2023	Response To Cytokine (GO:0034097)	0
Reactome_2022	Immune System R-HSA-168256	0
GO_Biological_Process_2023	Defense Response To Virus (GO:0051607)	0
GO_Biological_Process_2023	Defense Response To Symbiont (GO:0140546)	0
Reactome_2022	Neutrophil Degranulation R-HSA-6798695	0
Reactome_2022	Innate Immune System R-HSA-168249	0
Reactome_2022	Cytokine Signaling In Immune System R-HSA-1280215	0
Reactome_2022	Interferon Signaling R-HSA-913531	0
GO_Biological_Process_2023	Response To Interferon-Beta (GO:0035456)	0
Reactome_2022	Interferon Alpha/Beta Signaling R-HSA-909733	0
GO_Biological_Process_2023	Regulation Of Phagocytosis (GO:0050764)	0
GO_Biological_Process_2023	Negative Regulation Of Viral Process (GO:0048525)	0
GO_Biological_Process_2023	Cytokine-Mediated Signaling Pathway (GO:0019221)	0
Reactome_2022	ISG15 Antiviral Mechanism R-HSA-1169408	0
Reactome_2022	Antiviral Mechanism By IFN-stimulated Genes R-HSA-1169410	0

Table 13. Top enriched pathways among genes upregulated in IRC-High patient cells.

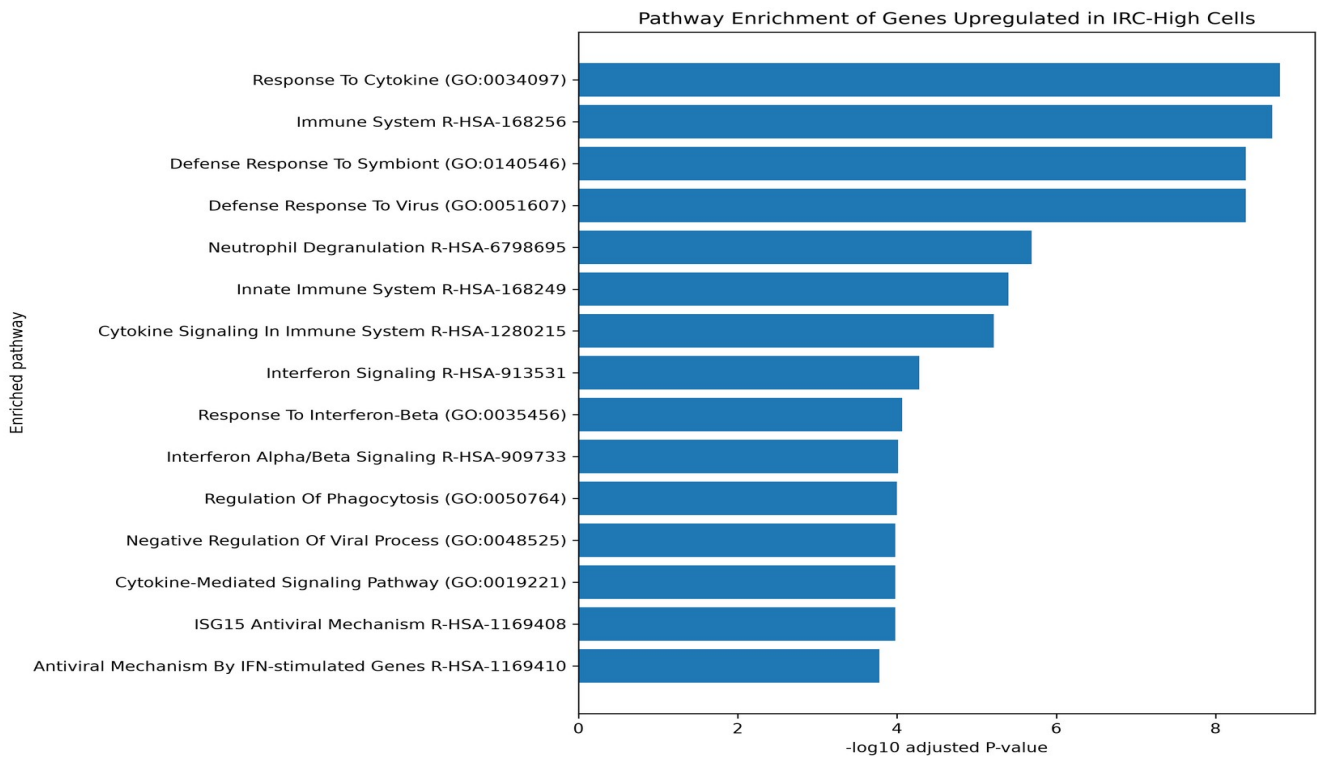


Figure 20. Top pathways enriched among genes upregulated in IRC-High patient cells.

3.9 Summary of Results

The sequential single-cell analysis generated a quality-controlled PBMC dataset following removal of low-quality cells, predicted doublets, and sparsely detected genes. Highly variable gene selection, PCA, UMAP, and Leiden clustering revealed 30 transcriptionally distinct cell clusters.

Canonical immune-marker expression demonstrated the presence of major PBMC lineages across the transcriptional landscape. Cell-level IRC scoring further revealed heterogeneity in Type I interferon-associated transcriptional activity across cells and samples.

Within patient-derived cells, median-based IRC stratification separated cells into IRC-High and IRC-Low transcriptional states. Differential-expression analysis identified genes associated with these states, and pathway enrichment was subsequently used to characterize biological processes represented among genes upregulated in IRC-High cells.

4. Discussion

4.1 Data Quality and Robustness of the Preprocessing Workflow

The QC figures and filtering summaries indicate that the preprocessing workflow effectively reduced low-quality and potentially artifactual cells while retaining the majority of the dataset. The raw merged dataset contained **101,020 cells and 36,601 genes**. After applying the basic QC thresholds of at least 300 detected genes per cell, mitochondrial transcript content below 15%, and hemoglobin transcript content below 5%, the dataset was reduced to **94,513 cells**, corresponding to removal of **6,507 cells**. Subsequent Scrublet filtering removed a further **7,065 predicted doublets**, leaving **87,448 cells**. Finally, filtering genes detected in fewer than five cells reduced the number of genes from **36,601 to 30,744**, while the cell count remained unchanged. Overall, **13,572 cells, or approximately 13.4% of the original dataset, were removed**, while **5,857 sparsely detected genes** were excluded.

The pre-filtering QC plots showed broad distributions in detected-gene counts, total transcript counts, mitochondrial percentage, and hemoglobin percentage across the ten samples. Several cells had more than **10,000 detected genes**, and total transcript counts reached above **100,000** in the most extreme cases. These high-complexity cells were of particular concern because unusually large transcriptomes can be consistent with multiplets and may disproportionately influence PCA, neighborhood construction, and clustering if retained. The pre-QC mitochondrial plots also contained cells with mitochondrial fractions extending to approximately **60–80%**, particularly among patient-derived samples, suggesting the presence of stressed, dying, or otherwise poor-quality cells. Hemoglobin content was generally low in most cells, although isolated pre-filtering outliers approached approximately **20%**.

The Scrublet diagnostic figures support the interpretation that a subset of high-complexity cells represented potential multiplets. Scrublet-predicted doublets were visibly shifted toward higher `n_genes_by_counts` values than predicted singlets and also showed a broader total-count distribution with a heavier upper tail. Importantly, the distributions of singlets and predicted doublets still overlapped substantially. This demonstrates why a simple upper threshold on total counts or detected genes would not have been sufficient for reliable doublet removal and supports the use of a dedicated doublet-detection method.

The sample-specific Scrublet results also showed that doublet burden was not uniform across libraries. The predicted doublet rate ranged from **3.39% in P5 to 10.76% in P2**. P2 contained the greatest number of predicted doublets (**1,482**), followed by P3 (**949**), HD2 (**834**), and P1 (**810**). Median Scrublet scores varied from **0.0487 in P8 to 0.0836 in P5**. These differences indicate that each sample had a distinct score distribution and support the decision to perform Scrublet detection separately for each library rather than applying one global threshold after concatenation.

Comparison of the pre- and post-filtering QC plots shows that the most extreme observations were substantially reduced after preprocessing. The upper tails of detected-gene counts and total counts were shortened, while all retained cells were restricted below the predefined **15% mitochondrial threshold**. Hemoglobin-associated transcript percentages were also reduced markedly, with the post-QC distributions almost entirely below approximately **1%**. Together, these changes indicate successful removal of the strongest low-quality and contamination-associated profiles.

The apparent increase in the visual size of some post-QC mitochondrial violins should, however, be interpreted carefully. The y-axis for mitochondrial percentage changes from approximately **0–80% before filtering to 0–15% after filtering**. Likewise, the hemoglobin axis changes from approximately **0–20% to below 1%**. The post-QC violins therefore represent a magnified view of the retained low-range values rather than an increase in mitochondrial or hemoglobin content. The correct comparison is the disappearance of the extreme high-value tails, not the apparent height or width of the violins across plots with different scales.

Residual sample-level heterogeneity remained after QC. In particular, patient samples such as **P5, P6, and P7** retained comparatively higher mitochondrial-content distributions, while **P5** showed lower detected-gene complexity than many other samples. At the donor-type level, patient-derived cells also retained a broader mitochondrial distribution than healthy-donor cells. These differences should not automatically be interpreted as failed QC, because mitochondrial content, transcriptional complexity, cell composition, activation state, and disease-associated biology can all vary between donors and samples.

Overall, the combined use of threshold-based filtering, sample-wise Scrublet detection, and low-frequency gene filtering produced a substantially cleaner final dataset of **87,448 cells and 30,744 genes**. The figures show that extreme low-quality and high-complexity observations were reduced while meaningful inter-sample variability was preserved, providing a more reliable basis for normalization, dimensionality reduction, clustering, and downstream IRC analysis.

Here is a polished **Section 4.2 Discussion** based on the UMAP, marker plots, and the three clustering CSVs you uploaded.

4.2 PBMC Transcriptional Structure and Immune-Lineage Composition

The UMAP embedding revealed a highly structured transcriptional landscape consisting of **30 Leiden clusters**. Cluster sizes varied substantially, from **5,826 cells in cluster 0** to only **75 cells in cluster 29**. The largest clusters were clusters 0, 1, 2, 3, and 4, each containing more than 5,000 cells, whereas clusters 26–29 represented much smaller populations, each accounting for less than approximately 0.5% of the dataset. This broad distribution suggests that the dataset contains both abundant PBMC populations and several comparatively rare transcriptional states.

The donor-type UMAP showed that healthy-donor and patient-derived cells were not distributed uniformly across the transcriptional space. Several clusters were overwhelmingly patient-derived,

including cluster 3 (**5,439 patient cells vs 3 healthy-donor cells**), cluster 7 (**4,441 vs 0**), cluster 8 (**3,975 vs 2**), cluster 11 (**3,559 vs 10**), cluster 16 (**2,761 vs 16**), and cluster 17 (**2,358 vs 3**). In contrast, some clusters were strongly enriched for healthy-donor cells, particularly cluster 5 (**4,587 healthy-donor vs 328 patient cells**) and cluster 12 (**3,441 vs 100**). These differences indicate that disease status is associated with substantial shifts in the abundance of particular transcriptional populations, although they should not be interpreted as purely disease-driven without considering sample composition.

The sample-level composition table showed that some clusters were also heavily dominated by individual samples. For example, cluster 7 was almost entirely composed of **P4 cells (4,437/4,441)**, cluster 8 was dominated by **P1 (3,944/3,977)**, cluster 3 by **P3 (5,252/5,442)**, cluster 11 by **P2 (3,396/3,569)**, and cluster 5 by **HD2 (4,303/4,915)**. Cluster 12 was composed predominantly of **HD1 and HD2 cells (1,406 and 2,035 cells, respectively)**. This strong sample enrichment is important because it suggests that at least some Leiden clusters may reflect donor- or sample-specific transcriptional states in addition to broad immune-cell identity. Therefore, the clustering structure should be interpreted as a combination of biological lineage, activation state, disease-associated variation, and possible residual inter-sample effects rather than as a direct one-to-one map of cell types.

The canonical marker dotplots nevertheless provided clear evidence that the major expected PBMC lineages were represented in the dataset. T-cell-associated markers such as **CD3D, IL7R, CCR7, CD4, and CD8A** were distributed across multiple clusters, consistent with the existence of several T-cell subpopulations rather than a single homogeneous T-cell cluster. The raw-matrix dotplot was especially informative because it recovered markers that were not necessarily present among the selected HVGs. For example, clusters 0, 1, 2, 16, 17, 19, 20, 21, 22, and 28 showed varying combinations of T-cell-associated markers, supporting substantial heterogeneity within the T-cell compartment.

NK-cell identity was supported by strong expression of **NKG7, GNLY, GZMB, and NCAM1** in a distinct subset of clusters. Clusters **4, 5, 13, and 18** showed particularly strong NK-associated expression, with high NKG7 and GNLY signal and, in several cases, elevated GZMB. The corresponding UMAP feature plots also showed NKG7 enrichment within distinct regions of the embedding, supporting the interpretation that NK-like transcriptional populations occupy specific parts of the UMAP rather than being diffusely distributed.

B-cell-associated markers were concentrated in a smaller number of clusters. **MS4A1** and **CD79A** were especially prominent in clusters **7, 14, and 25**, with the raw dotplot showing particularly strong MS4A1 expression in clusters 14 and 25. The MS4A1 UMAP similarly highlighted a discrete region of the embedding, consistent with a transcriptionally distinct B-cell compartment.

Monocyte-associated markers showed another distinct pattern. **CD14** and **LYZ** were enriched in multiple clusters, particularly clusters **3, 6, 8, 9, 11, 12, 23, and 24**, with cluster 24 showing especially strong LYZ expression. **FCGR3A**, which can mark non-classical monocytes, was more strongly expressed in a subset of clusters including 10 and 18. These patterns suggest that the monocyte compartment is subdivided into multiple transcriptional states, potentially including classical and non-classical populations.

Dendritic-cell-associated markers were less broadly distributed, as expected for a rarer PBMC lineage. **FCER1A** and **CST3** showed strongest enrichment in a limited subset of clusters, particularly around clusters **23, 24, 26, and 27**, with **CST3** also present in several monocyte-associated clusters. This partial overlap is biologically plausible because dendritic cells and monocytes can share myeloid-associated genes such as **CST3**. Therefore, assignment of DC identity should rely on the combined marker pattern rather than **CST3** alone.

The UMAP feature plots were consistent with the dotplots and helped visualize the spatial organization of the main lineages. **CD3D** and **CD4** were enriched across broad T-cell-dominated regions, **CD8A** highlighted a partially distinct T-cell region, **MS4A1** localized to a compact B-cell-associated area, **CD14** marked a separate myeloid-rich region, and **NKG7** highlighted NK/cytotoxic populations. The fact that these canonical markers map to spatially coherent UMAP regions supports the biological validity of the transcriptional structure identified by PCA, neighborhood construction, UMAP, and Leiden clustering.

Taken together, the clustering and marker-expression results support the presence of the major circulating PBMC lineages expected in the dataset, including CD4+ T cells, CD8+ T cells, NK cells, B cells, monocytes, and dendritic cells. However, the **30 Leiden clusters should not be interpreted as 30 distinct cell types**. Rather, the marker patterns indicate that several broad immune lineages are subdivided across multiple clusters, likely reflecting finer cell states, activation programs, and donor-specific transcriptional variation. The strong enrichment of several clusters for individual samples further indicates that sample identity remains an important contributor to the observed transcriptional structure.

4.3 Heterogeneity of Interferon Response Capacity

All six genes comprising the IRC signature—**BST2, EIF2AK2, ISG15, MX1, IFIT3, and IRF7**—were present in the full normalized expression matrix, allowing the IRC score to be calculated using the complete six-gene signature. Across all **87,448 cells**, IRC scores showed a broad distribution, ranging from **-0.381 to 2.189**, with an overall mean of **0.0045** and median of **-0.0364**. The interquartile range extended from **-0.178 to 0.114**, indicating that most cells had relatively modest IRC scores, while a smaller subset exhibited markedly elevated interferon-associated transcriptional activity.

The IRC-colored UMAP demonstrated that high IRC activity was not distributed uniformly across the transcriptional landscape. Most cells displayed low-to-moderate scores, whereas scattered regions contained cells with substantially higher IRC values. These high-score cells appeared within particular portions of the UMAP rather than across all clusters equally, suggesting that interferon-associated activity is linked to specific transcriptional states or immune-cell subsets. The existence of localized high-IRC regions is consistent with the idea that Type I interferon responsiveness varies according to cellular identity and activation state rather than functioning as a uniform property of all PBMCs.

Substantial heterogeneity was also evident across individual samples. The two healthy donors had very similar mean IRC scores, with **HD1 = -0.0349** and **HD2 = -0.0356**, and medians of **-0.0786** and

-0.0676, respectively. In contrast, the patient samples showed much wider variation. **P5** had the highest mean IRC score (**0.1696**) and the largest standard deviation (**0.4815**), while **P1** showed a similarly elevated mean (**0.1620**) and the highest patient median (**0.1268**). At the other extreme, **P4** exhibited the lowest mean IRC score (**-0.0846**) and median (**-0.1389**). The remaining patient samples showed intermediate profiles, including P2 with a mean of **0.0346**, P6 with **0.0060**, P7 with approximately **0.0000**, P3 with **-0.0247**, and P8 with **-0.0215**.

The sample-level violin plot reinforces these numerical differences. P1 displays a visibly broader distribution shifted toward higher IRC values, whereas P4 is concentrated more strongly at negative scores. P5 is particularly notable because its median remains slightly negative (**-0.0177**) despite having the highest mean IRC score. This discrepancy is explained by the strongly right-skewed distribution in P5, which contains a relatively small subset of cells with very high IRC activity. Indeed, P5 also contained the highest individual IRC score in the dataset (**2.189**). This pattern suggests that elevated IRC within a sample may arise from a specific subpopulation of highly interferon-responsive cells rather than from a uniform shift of the entire sample.

The donor-type violin plot similarly indicates greater heterogeneity among patient-derived cells. Healthy-donor cells are concentrated predominantly around slightly negative IRC values, whereas the patient distribution exhibits a broader positive tail extending beyond 2.0. However, the substantial overlap between healthy-donor and patient-derived cells indicates that IRC is not simply a binary property separating healthy and cancer samples. Rather, the stronger upper tail among patients appears to reflect the presence of particular high-IRC cell states within the patient population.

An important feature of these results is the distinction between **mean and median IRC values**. Several patient samples, particularly P5 and P2, had means that were noticeably higher than their medians, indicating right-skewed score distributions driven by subsets of highly responsive cells. This reinforces the single-cell nature of the IRC phenotype: an elevated average score for a sample does not necessarily imply that most cells in that sample are strongly interferon-responsive.

Overall, the IRC analysis revealed pronounced **cell-to-cell and sample-to-sample heterogeneity** in baseline Type I interferon-associated transcriptional activity. The presence of localized high-IRC regions on the UMAP, together with the strong differences in score distributions among individual patient samples, suggests that baseline interferon responsiveness is concentrated within particular cellular states rather than being uniformly elevated across the patient PBMC compartment. This heterogeneity provides the rationale for examining IRC-High and IRC-Low transcriptional states in subsequent differential-expression analyses.

4.4 Transcriptional Features of IRC-High and IRC-Low Cells

The IRC-High versus IRC-Low comparison was performed on **69,467 patient-derived cells**, divided almost exactly equally by the median IRC cutoff of **-0.02767**. This resulted in **34,734 IRC-High cells (50.0007%)** and **34,733 IRC-Low cells (49.9993%)**. The near-equal group sizes are expected because

the classification was based on the median patient-cell IRC score and provide a balanced cell-level comparison.

The volcano plots revealed a strongly asymmetric transcriptional pattern, with substantially more genes upregulated in IRC-High cells than downregulated. Using the predefined significance criteria of adjusted p-value < 0.05 and absolute log2 fold change > 0.5, **867 genes** were significant in total. Of these, **715 genes were upregulated in IRC-High cells**, whereas **152 genes were downregulated**. This predominance of positively regulated genes indicates that the IRC-High state is associated with a broad transcriptional activation program rather than simply a small shift in the six genes used to calculate the IRC score.

Importantly, the genes showing the strongest positive fold changes included all six members of the IRC signature. **IFIT3** showed the largest positive log2 fold change among the significant genes (**log2FC = 5.38**), followed by **MX1 (4.43)**, **EIF2AK2 (4.30)**, **IRF7 (4.11)**, **ISG15 (4.00)**, and **BST2 (3.61)**. This is internally consistent with the scoring framework, because cells classified as IRC-High are expected to show stronger expression of the interferon-response genes used to construct the score. However, the differential-expression pattern extended well beyond the original six-gene signature.

Several additional genes associated with interferon and antiviral responses were strongly enriched in IRC-High cells. These included **SIGLEC1 (log2FC = 2.51)**, **IFIT1 (2.43)**, **IFI44L (2.02)**, **CMPK2 (1.89)**, **OAS3 (1.82)**, **RSAD2 (1.79)**, and **OASL (1.75)**. The presence of multiple canonical interferon-stimulated genes outside the scoring set suggests that the IRC-High classification captures a broader Type I interferon-associated transcriptional state rather than merely reflecting mathematical separation based on the six input genes.

The labeled volcano plot further highlighted genes such as **STAT1**, **JAK2**, **EPSTI1**, **XAF1**, **TYMP**, **RNF213**, **AOAH**, **NEAT1**, and **SLC8A1** among the most statistically significant features of the IRC-High state. Several of these genes are directly or indirectly connected to cytokine signaling and interferon-responsive transcription. In particular, **STAT1** is a central transcription factor in interferon signaling, while **JAK2** participates in cytokine receptor signaling. Their enrichment supports the interpretation that IRC-High cells exhibit coordinated activation of interferon-associated signaling pathways rather than isolated overexpression of individual marker genes.

The downregulated side of the comparison was smaller and showed more modest fold changes. Among the strongest negatively associated genes were **CNR1 (log2FC = -1.28)**, **GABRB2 (-1.24)**, **ROR1 (-1.18)**, **MROH9 (-1.17)**, **GLRA3 (-1.15)**, **HRK (-1.11)**, **APOD (-1.07)**, **ABCB4 (-1.06)**, **CD200 (-1.05)**, and **PTPRG (-1.04)**. Compared with the pronounced positive shifts observed for interferon-stimulated genes, the IRC-Low-associated side of the volcano plot was less dominated by a single coherent transcriptional program.

A notable feature of the volcano plots is the presence of extremely small adjusted p-values, with several genes reaching values effectively reported as zero by the numerical output. This reflects the very large number of cells included in the Wilcoxon test and should be interpreted cautiously. Because thousands

of cells from the same patients are treated as individual observations, statistical significance can become extremely strong even for relatively modest expression differences. For this reason, the biological interpretation should emphasize **effect size and pathway coherence**, rather than p-value magnitude alone.

Overall, the differential-expression results indicate that IRC-High patient cells are characterized by a broad and coordinated interferon-responsive transcriptional program. The enrichment of genes such as **IFIT3, MX1, EIF2AK2, IRF7, ISG15, BST2, IFI44L, OAS3, RSAD2, OASL, STAT1, and JAK2** provides strong evidence that the IRC score reflects a biologically coherent Type I interferon-associated cell state. At the same time, because this comparison was performed at the **cell level rather than the patient level**, these results should be interpreted as exploratory transcriptional differences between cellular states rather than independent evidence of patient-level IRC phenotypes.

4.5 Biological Interpretation of Pathway Enrichment

Pathway enrichment analysis of the top 100 genes upregulated in IRC-High cells revealed a highly coherent immune and antiviral signature. The strongest enriched term was **Response to Cytokine (GO:0034097)**, with an adjusted p-value of 1.55×10^{-9} and an overlap of **12/125 genes**. This was followed by the broad Reactome **Immune System** pathway (**adjusted p = 1.94×10^{-9} ; 35/1943 genes**) and the GO terms **Defense Response to Virus** and **Defense Response to Symbiont**, both with adjusted p-values of 4.15×10^{-9} . These results indicate that the transcriptional differences associated with the IRC-High state are dominated by coordinated immune-defense and cytokine-response programs rather than isolated gene-level changes.

Several of the most significantly enriched pathways were directly related to Type I interferon biology. Reactome **Interferon Signaling** was significantly enriched (**adjusted p = 5.23×10^{-5}**), as was **Interferon Alpha/Beta Signaling (adjusted p = 9.67×10^{-5})**. The GO term **Response to Interferon-Beta** was also strongly enriched (**adjusted p = 8.65×10^{-5}**). Importantly, these pathways were supported by multiple interferon-responsive genes, including **BST2, STAT1, MX1, MX2, EIF2AK2, ISG15, XAF1, and TRIM25**. This convergence across Reactome and GO provides strong functional support for the interpretation that IRC-High cells represent a biologically coherent interferon-responsive transcriptional state.

The enrichment pattern also extended to antiviral effector functions. Reactome pathways such as **ISG15 Antiviral Mechanism (adjusted p = 1.06×10^{-4})** and **Antiviral Mechanism by IFN-Stimulated Genes (adjusted p = 1.68×10^{-4})** were among the top-ranked terms. Similarly, the GO term **Negative Regulation of Viral Process** was enriched (**adjusted p = 1.05×10^{-4}**). These pathways were driven by genes such as **STAT1, MX1, MX2, EIF2AK2, ISG15, BST2, IFI44L, and TRIM25**, which are well-established components of antiviral and interferon-stimulated transcriptional programs. The pathway-level results therefore reinforce the differential-expression findings and show that the IRC-High phenotype extends beyond the six genes used to calculate the score.

Broader innate immune processes were also represented. **Innate Immune System** enrichment was significant (**adjusted p** = 3.98×10^{-6} ; **21/1035 genes**), while **Cytokine Signaling in Immune System** was enriched with an adjusted p-value of 6.08×10^{-6} . Additional pathways included **Regulation of Phagocytosis**, **Neutrophil Degranulation**, and antigen-processing-related terms. These enrichments suggest that high IRC activity may be associated not only with interferon signaling itself, but also with broader activation of innate immune effector mechanisms and myeloid-associated functions.

The presence of **Neutrophil Degranulation** among the top pathways should be interpreted cautiously because the dataset consists of PBMCs rather than purified neutrophils. This enrichment likely reflects shared innate immune genes, such as **FCN1**, **CLEC12A**, **SLC11A1**, **CYBB**, **CTSS**, **CD36**, and **TLR2**, rather than direct evidence of a substantial neutrophil population. In the same way, broad terms such as **Immune System** are biologically consistent but less specific than the interferon-focused pathways.

Overall, the enrichment results strongly support the interpretation that IRC-High cells are characterized by an activated Type I interferon-associated immune program. The agreement between the DE analysis and multiple independent pathway databases is particularly important: genes identified individually as enriched in IRC-High cells converge at the pathway level on **cytokine signaling, interferon alpha/beta signaling, antiviral defense, and innate immune activation**. This functional coherence strengthens the conclusion that the IRC score captures a broader interferon-responsive cellular state rather than simply reflecting expression of the six genes used to construct the score. Because the pathway analysis was derived from a cell-level DE comparison, however, these findings should still be interpreted as exploratory transcriptional associations rather than independent patient-level pathway differences.

4.6 Comparison With the Original Study

The present analysis reproduces several important themes from the original study by Boukhaled et al., but it does so using a simplified scRNA-seq framework and therefore should not be interpreted as an exact replication of the authors' IRC methodology. The original study defined IFN-I response capacity by measuring the **change in expression of six interferon-stimulated proteins—BST2, PKR/EIF2AK2, ISG15, MX1, IFIT3, and IRF7—between IFN- β -stimulated and unstimulated peripheral immune cells**. A higher IRC therefore represented a greater inducible response to IFN-I stimulation. In contrast, the present analysis calculated a Scanpy module score from the basal scRNA-seq expression of the same six genes in `adata.raw`. Thus, the score used here reflects a **baseline interferon-associated transcriptional state**, rather than the experimentally measured inducible IFN-I responsiveness used in the original paper.

Despite this methodological difference, several findings were biologically concordant with the study. Boukhaled et al. emphasized substantial inter-patient heterogeneity in IFN-I responsiveness and showed that total PBMC-level IRC was insufficient to explain clinical outcome; the strongest associations emerged specifically within CD4 and CD8 effector T-cell populations. Similarly, the present analysis identified marked heterogeneity in IRC scores across individual cells and samples,

with localized high-IRC regions on the UMAP rather than a uniform shift across the entire PBMC compartment. This supports the broader conclusion that IFN-I-associated immune states are heterogeneous and cell-context dependent.

The original study's central clinical result was counterintuitive: **lower pre-therapy IFN-I responsiveness in peripheral CD4 and CD8 effector T cells was associated with improved survival after anti-PD1 therapy, whereas high responsiveness was associated with progression and shorter survival.** The discovery and validation cohorts consistently showed longer overall survival in patients with low pre-therapy CD4 Teff IRC, and similar trends were observed for CD8 Teff cells. The current analysis cannot directly test this survival relationship because patients were not stratified according to clinical outcome and the IRC-High versus IRC-Low grouping was performed at the **cell level**, not at the patient level. Therefore, the present results support the existence of distinct interferon-associated transcriptional states but do not independently validate the paper's prognostic conclusion.

There is also an important difference in the transcriptional interpretation of high IRC. In the original study, baseline expression of the six IRC-associated ISGs was reported to be similarly low in high- and low-IRC CD4 Teff groups. The authors argued that the distinction between high and low IRC did not result from constitutively elevated ISG expression, but instead reflected a pre-existing cellular physiology that determined how strongly cells could respond when subsequently exposed to IFN-I. By contrast, in the current analysis the same six genes were used directly to compute the basal IRC module score. As expected, IRC-High cells therefore showed strong upregulation of IFIT3, MX1, EIF2AK2, IRF7, ISG15, and BST2, together with additional interferon-stimulated genes such as IFIT1, IFI44L, OAS3, RSAD2, and OASL. This difference is a consequence of the scoring strategy and means that the present "IRC-High" state is closer to a **basal interferon-stimulated transcriptional phenotype** than to the experimentally measured response capacity defined in the original study.

Nevertheless, the broader functional patterns show meaningful agreement with the biological framework of the paper. The current pathway-enrichment analysis identified cytokine response, interferon signaling, interferon alpha/beta signaling, response to interferon-beta, antiviral defense, innate immune activation, and ISG15-mediated antiviral mechanisms among the strongest pathways enriched in IRC-High cells. These results are consistent with the central role of IFN-I-responsive programs described by Boukhaled et al. The original study further showed that IRC-high CD4 Teff cells were enriched for transcriptional programs associated with T follicular helper-cell differentiation and T-cell dysfunction, whereas IRC-low cells were enriched for memory-associated programs; high-IRC CD8 Teff cells also showed increased exhaustion signatures. The current analysis did not specifically isolate CD4 or CD8 Teff cells for IRC-high versus IRC-low DE, so those lineage-specific dysfunction and memory programs cannot yet be directly compared.

Another major finding of the original study was that IFN-I responsive states were associated with distinct epigenetic and transcriptional programs already present before therapy. Multiome analysis demonstrated differences in both gene expression and chromatin accessibility between high- and low-IRC CD4 Teff cells, suggesting that responsiveness to IFN-I is epigenetically programmed rather than determined solely by basal ISG abundance. The authors further showed that high-IRC cells were

associated with transcription-factor networks linked to T-cell dysfunction, including STAT1, FLI1, RUNX1 and NFAT-related programs, whereas low-IRC cells showed stronger AP-1-associated activity. The current analysis observed enrichment of **STAT1 and JAK2** among the most significant IRC-High DE genes, which is directionally consistent with stronger interferon-associated signaling, but without scATAC-seq or regulon-level analysis the epigenetic mechanism proposed in the paper cannot be evaluated directly.

The paper also demonstrated that peripheral IRC was not simply a reflection of tumor inflammation. Pre-therapy CD4 and CD8 Teff IRC did not correlate with baseline tumor-intrinsic inflammatory scores, supporting a distinction between systemic immune responsiveness and the inflammatory state of the tumor microenvironment. The current dataset contains PBMC transcriptomic data only and therefore cannot investigate this peripheral–tumor disconnect.

Overall, the present analysis provides a **partial biological reproduction** of the original study. It successfully recovers the same six-gene IFN-I signature, demonstrates strong cell- and sample-level heterogeneity in interferon-associated states, identifies a broad IRC-High transcriptional program dominated by interferon, cytokine and antiviral-response genes, and recovers pathway-level signatures consistent with heightened IFN-I activity. However, it differs fundamentally from the original work in that IRC was inferred from basal transcript abundance rather than from IFN- β -induced changes, the analysis was performed across patient cells rather than specifically within CD4 and CD8 effector T-cell populations, and no patient-level survival or treatment-response analysis was performed. Consequently, the current findings are best interpreted as supporting the **biological coherence of interferon-associated cellular states**, while the original study provides the stronger evidence linking experimentally measured pre-therapy IFN-I response capacity to anti-PD1 treatment outcome.

4.7 Limitations

Several limitations should be considered when interpreting the results.

First, IRC-High and IRC-Low classification was performed at the **cell level** using the median IRC score across patient-derived cells. Entire patients were therefore not independently classified as IRC-High or IRC-Low.

Second, the Wilcoxon differential-expression analysis treats individual cells as observations even though multiple cells originate from the same biological donor. This introduces potential pseudoreplication and means that the reported p-values should not be interpreted as equivalent to patient-level statistical evidence.

A replicate-aware approach, such as pseudobulk differential expression performed at the patient level, would provide stronger inference if sufficient biological replicates and appropriate clinical group labels were available.

Third, Leiden clustering and marker-based lineage assessment provide a broad description of PBMC populations but do not constitute a fully validated cell-type annotation strategy. More detailed

annotation using additional markers, reference mapping, or automated annotation approaches could further resolve immune-cell subtypes.

Finally, pathway-enrichment results depend on the selected DE thresholds and the top 100 upregulated genes supplied to Enrichr. These enrichment results should therefore be interpreted as exploratory summaries of the transcriptional programs associated with the IRC-High state.

4.8 Overall Interpretation

Taken together, the analysis identified substantial heterogeneity in baseline Type I interferon-associated transcriptional activity across PBMCs. IRC-High patient cells exhibited a distinct transcriptional profile and corresponding pathway-level differences relative to IRC-Low cells.

These observations are consistent with the broader hypothesis that baseline interferon responsiveness represents an important component of immune-cell state and may influence how circulating immune populations respond to immune-checkpoint blockade.

The findings therefore provide a biologically coherent computational reproduction of the central concept investigated in the original study, while also highlighting the need for patient-level, replicate-aware validation of the observed IRC-associated transcriptional differences.

4.7 Limitations

Several limitations should be considered when interpreting the present findings.

First, the IRC score used in this analysis is not identical to the IRC definition used in the original study. Boukhalel et al. calculated IRC from the change in expression of six interferon-stimulated proteins after IFN- β stimulation relative to unstimulated cells. In contrast, the present analysis calculated a Scanpy module score from baseline scRNA-seq expression of the corresponding six genes. The resulting score therefore reflects a basal interferon-associated transcriptional state rather than experimentally measured IFN-I response capacity. This difference is especially important because the original study reported similarly low baseline ISG expression in high- and low-IRC CD4 Teff groups, suggesting that the biological distinction was primarily in inducible responsiveness rather than constitutive ISG expression.

Second, the IRC-High and IRC-Low groups were defined at the **cell level**, not at the patient level. Patient-derived cells were divided according to the median IRC score across all patient cells, meaning that individual cells from the same patient could belong to either IRC group. This differs substantially from the original study, where patients were classified according to IRC measured within specific immune subsets.

Third, the differential-expression analysis treated individual cells as observations even though many cells originated from the same patient. This introduces a risk of **pseudoreplication**, because cells from the same biological donor are not statistically independent. The extremely small adjusted p-values observed for many genes are partly influenced by the very large number of cells included in the Wilcoxon test. For this reason, effect sizes and biological coherence are more informative than p-value magnitude alone. A replicate-aware method such as pseudobulk differential expression would provide stronger patient-level inference.

Fourth, the analysis was performed across all patient-derived PBMCs rather than specifically within CD4 or CD8 effector T-cell populations. This is an important difference from the original study, where the strongest association between IRC and survival was observed in CD4 and CD8 Teff cells. Consequently, lineage-specific programs such as T-cell exhaustion, follicular-helper-like states, or memory-associated transcription could not be evaluated with the same specificity.

Fifth, Leiden clustering identified 30 transcriptional clusters, but these clusters were not subjected to a fully validated annotation pipeline. Canonical marker expression supported the presence of major PBMC lineages, but several clusters were strongly enriched for individual samples. This suggests that some clusters may reflect donor-specific transcriptional states, activation programs, or residual technical variation in addition to cell identity.

Sixth, pathway enrichment depended on predefined thresholds and on the top 100 genes upregulated in IRC-High cells. Different significance thresholds, ranking strategies, or enrichment methods could produce somewhat different pathway results. The ORA findings should therefore be viewed as functional summaries rather than definitive evidence of pathway activation.

Seventh, the present dataset did not include patient survival, treatment-response categories, tumor transcriptomic data, or paired pre- and post-treatment measurements. Therefore, the central clinical claim of the original study—that low pre-therapy Teff-cell IRC predicts improved response and survival after anti-PD1 therapy—could not be directly tested.

Finally, the original study included multi-omic analyses integrating scRNA-seq and scATAC-seq, allowing investigation of epigenetic mechanisms underlying IFN-I responsiveness. The present analysis was transcriptomic only, so differences in chromatin accessibility and transcription-factor regulation could not be assessed directly.

4.8 Overall Interpretation

Taken together, the present analysis demonstrates that baseline interferon-associated transcriptional activity is highly heterogeneous across PBMCs. IRC scores varied substantially both between individual cells and between samples, with higher scores concentrated within specific regions of the transcriptional landscape rather than being uniformly distributed across the dataset.

The differential-expression analysis further showed that IRC-High cells were characterized by a broad interferon-responsive program extending beyond the six genes used to calculate the score. Strongly upregulated genes included IFIT3, MX1, EIF2AK2, IRF7, ISG15, BST2, IFIT1, IFI44L, OAS3, RSAD2, OASL, STAT1, and JAK2. This pattern suggests that the IRC module score captures a coherent IFN-I-associated transcriptional state rather than merely reflecting variation in a small number of marker genes.

Pathway-enrichment results reinforced this interpretation. The strongest enriched pathways included cytokine response, interferon signaling, interferon alpha/beta signaling, response to interferon-beta, antiviral defense, innate immune activation, and ISG15-associated antiviral mechanisms. The

agreement between gene-level and pathway-level results supports the biological validity of the observed IRC-High state.

The findings are partially consistent with the framework proposed by Boukhaled et al., in which IFN-I responsiveness reflects an important pre-existing immune-cell state. However, the biological meaning is not identical because the present score represents basal transcriptional activity rather than induced IFN-I responsiveness. In the original paper, high IRC was associated with dysfunctional effector T-cell states and poorer outcome after PD1 blockade, while low IRC was associated with memory-related programs and improved survival.

Therefore, the current results support the broader concept that interferon-responsive states are heterogeneous and transcriptionally distinct, but they do not independently reproduce the prognostic component of the original study. The analysis is best interpreted as a transcriptomic exploration of IFN-I-associated cellular states in PBMCs and as a partial computational reproduction of the biological themes described in the original work.

5. Conclusion

This study applied a complete single-cell RNA-sequencing workflow to PBMCs from two healthy donors and eight cancer patients and generated a quality-controlled dataset of **87,448 cells and 30,744 genes**. Following normalization, highly variable gene selection, dimensionality reduction, and Leiden clustering, 30 transcriptional clusters were identified. Canonical marker analysis demonstrated the presence of the major expected PBMC lineages, including CD4⁺ T cells, CD8⁺ T cells, NK cells, B cells, monocytes, and dendritic cells.

A six-gene interferon-associated module score based on BST2, EIF2AK2, ISG15, MX1, IFIT3, and IRF7 revealed marked heterogeneity in baseline Type I interferon-associated transcriptional activity across cells and samples. IRC-High patient cells displayed a distinct transcriptional profile characterized by broad upregulation of interferon-stimulated genes and enrichment of pathways related to cytokine signaling, Type I interferon signaling, antiviral defense, and innate immune activation.

These findings demonstrate that the six-gene signature identifies a coherent interferon-associated transcriptional state within the PBMC compartment. They are consistent with the broader biological premise of the original study that IFN-I-responsive states represent meaningful components of immune-cell physiology.

However, the present analysis differs fundamentally from the original experimental design. The original IRC measured inducible IFN-I responsiveness following IFN- β stimulation, whereas the current score was derived from baseline RNA expression. In addition, the present stratification and differential-expression analyses were performed at the cell level and did not evaluate independent patient-level clinical outcomes.

Accordingly, the analysis should be considered a **partial computational reproduction and extension** of the original study rather than an exact replication. Future analyses incorporating stimulated and unstimulated samples, cell-type-specific IRC calculation, replicate-aware differential expression,

patient-level stratification, and clinical outcome data would provide a more direct test of the relationship between pre-therapy IFN-I responsiveness and anti-PD1 treatment response.

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7. Appendix: AI Usage Disclosure

7.1 Tool and Model Disclosure

- **AI Tool:** ChatGPT, Gemini Notebook (assisted by the underlying reasoning and development agent).
- **Core Role:** Assisted in report structuring, methods documentation, biological results translation, and critical methodology comparison between student and paper workflows. Commenting on codes and generate codes for volcano plots and bar plot for pathway enrichment as there was no code in the guideline for this.

7.3 Prompt and Purpose Log

Purpose of Submitted Prompt	Functionality
Give me code blocks of the script with comments	To understand the code clearly
Tell me what to do to run to generate the bar plot and volcano plot	To generate Figure 5
Generate the submittable paper structure with my findings	To structure the Methods and Results sections of the paper
Discuss the findings based on the given rules	To develop the Discussion section

7.3 Student Verification Statement

All AI-assisted outputs, descriptions, and structural elements were independently reviewed, cross-referenced with the source documents (01_data_loading_qc.docx through 05_pathway_enrichment.docx, what I did. docx, project3_guide.pdf, and paper.pdf), and validated for factual accuracy. Parameter choices and statistical interpretations have been fully validated, and we assume complete academic authorship over this report.